

A preliminary design method for jacket support structures of high-capacity offshore wind turbines

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ABSTRACT

Preliminary designs are essential for optimizing jacket structures used to support offshore wind farms in transitional water depths and analyzing the integrated dynamic response of jacket-offshore wind turbines (JOWTs). This study proposes a systematic jacket design framework for offshore wind turbines (OWTs), achieving layer-specific member sizing with high computational efficiency (results are achieved typically within 10 min post-data preparation). First, governing aerodynamic and hydrodynamic load cases (LCs) for jacket and pile design are identified according to relevant standard requirements. Then, a top-down, layered design methodology incorporating the latest hydrodynamic loads in the loop is introduced. The overall jacket geometry is determined based on site parameters and engineering experience; initial member cross-sections are set to meet the natural frequency requirement; and member dimensions are finalized by iterative calculations. Subsequently, equivalent stiffness for tapered jackets is derived through equivalent horizontal displacement at the transition piece, facilitating verification of compliance with system natural frequency and tower-top deflection constraints. Finally, the validation against jacket designs from the Upwind and INNWIND projects confirms the reliability of the proposed method. A MATLAB computational tool implementing this methodology has been developed and made publicly available on GitHub. Collectively, the proposed methodology and tool provide direct technical support for integrated offshore wind turbine analysis and design.

1. Introduction

The offshore wind power has become a pivotal renewable energy solution addressing environmental concerns and insatiable global energy demands [1,2]. The primary foundation types for offshore wind turbines (OWTs) include gravity base, monopile, tripod, jacket, and floating structures [3,4]. While monopiles currently dominate installations, their cost and technical constraints restrict deployment to shallow waters (≤ 30 m depth) [5,6]. As shallow-water wind resources approach development saturation, the wind industry focus has shifted toward transitional depths (30–60 m). Consequently, the jacket foundations are increasingly deployed in this range due to advantages including lighter structural mass, enhanced stiffness, and high strength [7].

Throughout their lifespan, jacket offshore wind turbines (JOWTs) endure stochastic wind-wave-current loads—the governing loads

requiring significant attention in support structure design. Kim and Lee [8] demonstrated that wind speed uncertainty dominates dynamic responses of JOWTs via reliability analysis. Shittu et al. [9] quantified hydrodynamic loads as critical drivers of serviceability limit state (SLS) reliability using stochastic finite element analysis. Dong et al. [10] identified wind loads as the primary contributor to cumulative fatigue damage, with wave loads and wind-wave interactions exerting secondary influences. Under extreme marine conditions, the structural integrity of JOWTs is significantly influenced by diverse loads and effects. For instance, Kim and Manuel [11] identified ocean currents critically influence structural behavior of JOWTs during Hurricane Sandy. Therefore, the design of JOWT support structures must comprehensively account for wind, wave, and current loads, as well as SSI across load cases.

Unlike axisymmetric monopiles, jacket geometry induces load-direction-dependent structural responses. Han et al. [12] investigated

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multiaxial wind load-induced fatigue damage in JOWTs considering load correlations. Romero-Sánchez and Padrón [13,14] proposed a multidirectional coupled seismic-wind analysis framework and quantified seismic vulnerability of JOWTs across operational modes (power production, emergency shutdown, parked). For long-period structures, careful consideration of soil-structure interaction (SSI) effects is also required. Abhinav and Saha [15] highlighted the critical role of SSI under extreme wave loading for JOWTs. Natarajan and Madabhushi [16] underscore the risk of soil liquefaction-induced excessive foundation settlements for JOWTs during earthquakes. James and Halder [17] demonstrated that the pile length-to-diameter ratio critically controls mudline rotations for 10-MW JOWT. Bai et al. [18] warned that scour substantially increase seismic vulnerability of JOWTs. Thus, designing JOWT support structures necessitates consideration of load directionality effects and seabed SSI.

The complexity inherent in the operating environment of JOWTs necessitates prioritizing the identification of governing jacket design parameters and the development of applicable SSI models for engineering design. Zhang et al. [19] identified critical structural design parameters and established an optimal balance between specifications and requirements for OWT jacket foundations using Monte Carlo simulation. Qian et al. [20] developed geometric parameter prediction models integrating rated power and water depth into a data-driven conceptual design framework for jacket foundations. Jalbi et al. [21] investigated anti-resonance design criteria for JOWTs, concluding that inadequate foundation vertical stiffness can induce rocking vibration modes. Antoniou et al. [22] proposed an enhanced Winkler-based model for efficient seismic design of JOWTs in clay. Collectively, these studies underscore that preliminary design must address structural analysis of key governing factors.

The fatigue design poses a critical challenge for OWT jacket due to continuous exposure to wind, waves, and currents. Recognizing the multitude of dynamic loads, Schorbach et al. [23] analyzed governing factors for jacket fatigue design, identifying wind loads as the dominant contributor. Addressing this, Ju et al. [24] developed a parallel fatigue design framework for OWT jackets, identifying power production faults at 15 m/s as the primary fatigue source. To enhance jacket structure fatigue life without cost escalation, Ju [25] proposed a yaw system stiffness and damping tuning method for JOWTs. Moreover, with increasing artificial intelligence (AI) applications in engineering, data scarcity and imbalance pose challenge to offshore jacket design. Panagiotou et al. [26] proposed a generative AI framework integrating probabilistic data synthesis and multi-objective optimization to improve surrogate model accuracy. These advances in fatigue design methodologies and tools facilitate increasingly streamlined and refined jacket foundation design.

The support structures of JOWTs are typically designed by specialized entities: the tower by turbine manufacturers, and the jacket (including foundations) by engineering contractors. Consequently, the integrated design optimization of the support structure is crucial for reducing the levelized cost of offshore wind power. Häfele et al. [27] optimized jacket structures for OWTs considering geometry, cost, and code compliance, while Feng et al. [28] proposed shape optimization for nodal coordinates and cross-sections. However, JOWT support structures feature numerous components, complex loading conditions, and multiple constraints. Schafhirt et al. [29] demonstrated that conventional two-stage local optimization approaches exhibit inherent limitations for OWT jackets. Compounding this, JOWTs undergo complex dynamic interactions with wind-wave fields and seabed soils during operation. Sandal et al. [30] developed an integrated optimization framework for OWT jackets and piles incorporating soil-pile interaction. Zhou et al. [31] proposed topology optimization within expanding design domains. As conventional methods require intensive computation, Zheng et al. [32] developed a Kriging and genetic algorithms-driven framework. Oest et al. [33] compared gradient-based methods, emphasizing the need for stable initial parameters for reliable

outcomes. Thus, initial parameters significantly influence optimization efficacy.

The integrated design optimization and dynamic response of JOWTs necessitates greater emphasis on preliminary jacket and pile design. Jalbi and Bhattacharya (2019) [34] proposed a concept design method for offshore wind turbine jacket structures, incorporating multidisciplinary considerations such as structural response, geotechnical behavior, and hydrodynamics. This methodology is suitable for designing jackets with uniform cross-section members and has been widely cited. However, in engineering practice, variable cross-section members are frequently adopted, rendering this approach inapplicable. To effectively enable the preliminary design and analysis of jacket structures with variable cross-sections, which offer significant economic benefits, an extension of the Jalbi-Bhattacharya method is essential. Consequently, this study introduces a novel integrated framework featuring: (1) a variable-cross-section jacket design method; and (2) equivalent stiffness models for dynamic analysis of variable-section jackets. The paper is structured as follows: Section 2 details the methodology, including load combination, equivalent stiffness modeling, and system frequency prediction; Section 3 validates the approach via case studies; and Section 4 presents conclusions and future research directions. An open-source MATLAB toolkit implementing this methodology is provided on GitHub. This resource, together with the proposed framework, provides practical support for integrated OWT analysis and design workflows.

2. Model and method

2.1. Loading cases

The coordinate system depicted in Fig. 1 defines the global position and orientation of structure, wind and wave field. The origin is positioned at the intersection of the mean sea level (MSL) and the undeformed tower axis. The x-axis points to east; the z-axis coincides with the undeformed tower axis (positive upward); and the y-axis completes the right-handed triad via $\hat{y} = \hat{z} \times \hat{x}$ (where $\hat{x}$, $\hat{y}$ and $\hat{z}$ are the unit vector along respective axes).

The design of OWT support structures requires the incorporation of both normal and extreme marine environmental conditions to ensure structural integrity. While international standards (e.g., IEC 61400-3-1 [35] and DNV GL-ST-0126 [36]) specify hundreds of load cases for OWTs design, empirical evidence indicates that the substructure and foundation configurations are primarily governed by a critical subset.

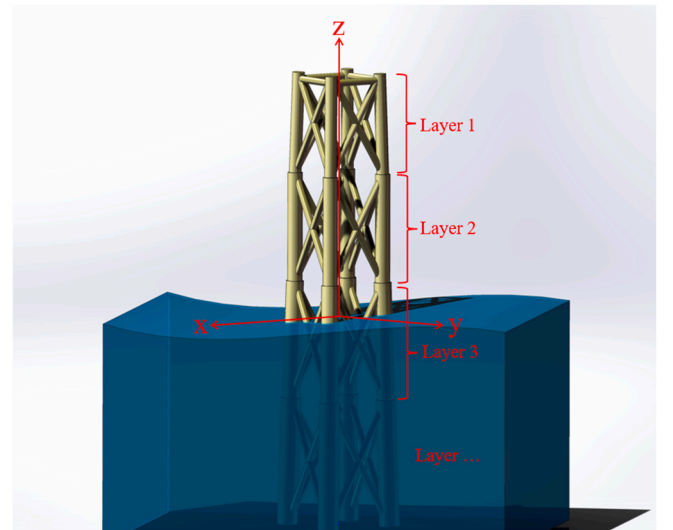


Fig. 1. Global coordinate system.

This study adopts the governing load cases in Table 1 for preliminary design of fixed-bottom OWT substructures according to IEC 61400-3-1 [35]. In Table 1, U_{in} , U_{out} , and U_R are the cut-in, cut-out and rated wind speed of wind turbines, respectively; U_{ref} is the reference wind speed of wind turbine; $\bar{U}_{hub}$ is the 10 min mean wind speed of hub-height for wind turbines; H_s is the significant wave height of the sea; H_{s1} and H_{s50} are the significant wave height with return period of 1-year and 50-year, respectively.

To guarantee structural integrity across varying operational regimes, the OWT support structure design must account for both the behavior of the turbine control system and environmental loading conditions. Current design standards categorize load cases into two primary groups: running states and parked conditions. This section summarizes load cases under these conditions.

2.1.1. Running condition of OWTs

Under the running conditions, wind field and sea state models associated with the ultimate limit state (ULS) of fixed-bottom OWTs include the extreme turbulence model (ETM), extreme operating gust (EOG), and normal sea state (NSS).

2.1.1.1. Extreme turbulence model. The wind velocity under turbulent is decomposed into a mean component $\bar{U}$ and stochastic components u , as mathematically represented by Eq. (1):

$$U = \bar{U} + u \quad (1)$$

where, $\bar{U}$ is the mean wind speed; u represents the fluctuating wind speed component. The load case 1.3 corresponds to normal power generation with hub-height mean wind speed between the cut-in and cut-out values, subjected to ETM turbulence.

According to IEC 61400-1 [37], the standard deviation of turbulence wind speed for the ETM, $\sigma_{U,ETM}$ is defined as:

$$\sigma_{U,ETM} = c I_{ref} \left[0.072 \left(\frac{\bar{U}_{avg}}{c} + 3 \right) \left(\frac{\bar{U}_{hub}}{c} - 4 \right) + 10 \right] \quad (2)$$

where, the parameter c is 2 m/s; I_{ref} is the reference turbulence intensity with a mean wind speed of 15 m/s at the hub height; $\bar{U}_{avg}$ is the long-term average wind speed at the hub height. For conservatism, both $\bar{U}_{hub}$ and $\bar{U}_{avg}$ are set to the rated wind speed of wind turbine. The standard deviation $\sigma_{U,ETM,f>1p}$, which quantifies turbulence components in frequency bands exceeding the 1P frequency, is determined through the following relationship:

$$\sigma_{U,ETM,f>1p} = \sqrt{\int_{f_{1p,max}}^{\infty} S_{uu}(f) df} = \sigma_{U,ETM} \sqrt{\frac{1}{\left(\frac{6L_K f_{1p,max}}{U_R} + 1 \right)^{\frac{2}{3}}}} \quad (3)$$

where, $S_{uu}(f)$ is the Kaimal wind spectrum model; f is the frequency; L_K is the integral length scale, empirically defined as 340.2 m for offshore environments according to DNV-OS-J101 [38] in this study; $f_{1p,max}$ represents the upper limit of the 1P frequency band. Under the

assumption of normally distribution and a 95% confidence interval, the fluctuating wind speed component of the ETM, u_{ETM} , is formulated as follows:

$$u_{ETM} = 2 \cdot \sigma_{U,ETM,f>1p} \quad (4)$$

2.1.1.2. Extreme operating gust. For the extreme operating gust (EOG) model, the wind turbine is in the running state, with the mean wind speed at hub height set to the rated wind speed. The gust magnitude at hub height, U_{gust} , is determined by the following equations:

$$U_{gust} = \min \left\{ 1.35(U_{e1} - \bar{U}_{hub}); 3.3 \left(\frac{\sigma_1}{1 + 0.1 \left(\frac{D}{\Lambda_1} \right)} \right) \right\} \quad (5)$$

$$\sigma_1 = I_{ref}(0.75\bar{U}_{hub} + b), b = 5.6 \text{ m} \quad (6)$$

where, U_{e1} is the extreme wind speed with a return period of 1 year; σ_1 is turbulence standard deviation; D denotes the rotor diameter; Λ_1 is the turbulence scale parameter and takes values: 0.7z if $z \leq 60$ m, and 42 if $z \geq 60$ m.

The time history of gust wind speed is mathematically described by Eq. (7):

$$U(z, t) = \begin{cases} \bar{U}(z) - 0.37U_{gust} \sin\left(\frac{3\pi t}{T_0}\right) \left(1 - \cos\left(\frac{2\pi t}{T_0}\right)\right) & 0 \leq t \leq T_0 \\ \bar{U}(z) & \text{otherwise} \end{cases} \quad (7)$$

where, $T_0 = 10.5$ s is the wave period; $\pi = 3.14$ is the value of the mathematical constant used; $\bar{U}(z)$ is the 10 min mean wind speed at height z and be determined by Eq. (8):

$$\bar{U}(z) = \bar{U}_{hub} \left(\frac{z}{z_{hub}} \right)^{\alpha_1} \quad (8)$$

where, α_1 is the wind shear exponent; z_{hub} is the hub height above the MSL, aligned with the z -coordinate of the hub center.

2.1.1.3. Normal sea state. The cross-sectional design of OWT jacket components requires appropriate wave theories to determine water particle kinematics. As shown in Fig. 2, the linear wave theory provides sufficient accuracy (<10% error) under deep-water conditions [39]. Thus, the Airy wave theory is adopted in this study to mathematically describe wave-induced fluid motions.

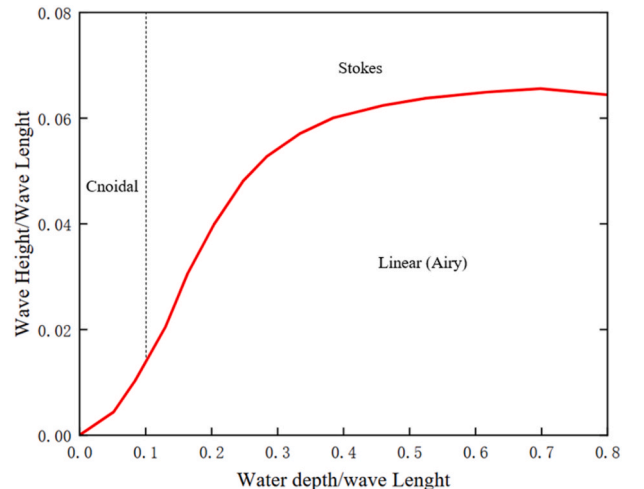


Fig. 2. Applicability range of linear wave theory [39].

Table 1
Design loading cases of OWT jacket support structure.

Design situation	DLC	Wind condition	Waves
Power production	1.3	ETM, $U_{in} < \bar{U}_{hub} < U_{out}$	NSS, $H_s = E[H_s \bar{U}_{hub}]$
Power production plus occurrence of fault	2.3	EOG, $\bar{U}_{hub} = U_R \pm \frac{2m}{s}$ and U_{out}	NSS, $H_s = E[H_s \bar{U}_{hub}]$
Parked (standing still or idling)	6.1	EWM Turbulent wind model, $\bar{U}_{hub} = U_{ref}$	ESS, $H_s = H_{s50}$
Parked (standing still or idling)	6.3	EWM Turbulent wind model, $\bar{U}_{hub} = U_1$	ESS, $H_s = H_{s1}$

Assuming sinusoidal waves (Fig. 3), the free surface elevation η is:

$$\eta = \frac{H}{2} \cos(kx - \omega_0 t) \quad (9)$$

where, H is the wave height; L is the wavelength; ω_0 is the wave frequency and can be computed via the dispersion equation (Eq. (10)) [40]:

$$k = g \cdot k \cdot \tanh(kS) \quad (10)$$

where, $g = 9.8 \text{ m/s}^2$ is the gravitational acceleration; S is the mean water depth; k is the wave number, which is a function of the wave length L , and is related to the wave frequency given by Eq. (10).

At time t and position (x, z) , the horizontal and vertical velocity components of water particles induced by the wave are:

$$\begin{aligned} u_{\text{wav},x} &= \frac{\omega_0 H \cosh(k(z+S))}{2 \sinh(kS)} \cos(kx - \omega_0 t) \\ u_{\text{wav},z} &= \frac{\omega_0 H \sinh(k(z+S))}{2 \sinh(kS)} \sin(kx - \omega_0 t) \end{aligned} \quad (11)$$

where, $u_{\text{wav},x}$ and $u_{\text{wav},z}$ are velocity components of water particles in x -axis and z -axis, respectively.

The ocean currents originate from sources including wind stress, tidal forcing, and thermohaline circulation, exhibiting pronounced variability in velocity profiles. DNV-RP-C205 [41] characterizes current as a depth-dependent steady-state flow. Specifically, the depth-resolved current velocity vector $u_{\text{cur}}(z)$ is formulated as a linear superposition of surface-generated and subsurface current constituents:

$$u_{\text{cur}}(z) = u_{\text{c},n}(z) + u_{\text{c},s}(z) \quad (12)$$

where, $u_{\text{c},n}(z)$ is the surface current velocity vector; $u_{\text{c},s}(z)$ is the subsurface current velocity vector.

The surface currents are predominantly wind-driven. The vertical velocity profile of these currents is commonly modeled as exhibiting an exponential decay with depth, reaching negligible values at depths exceeding 20 m. Through the upper 20 m oceanic layer, the depth-dependent surface current velocity profile, denoted as $u_{\text{c},n}(z)$, can be mathematically described by Eq. (13):

$$u_{\text{c},n}(z) = u_{\text{c},w}(0) \left(1 + \frac{z}{20}\right), \quad -20 \leq z \leq 0 \quad (13)$$

where, $u_{\text{c},w}(0)$ is the wind-induced current velocity magnitude at the sea surface. The subsurface current velocity profile $u_{\text{c},s}(z)$ is described by the exponential decay model in Eq. (14):

$$u_{\text{c},s}(z) = u_{\text{c},s}(0) \left(\frac{S+z}{S}\right)^\beta, \quad z \leq 0 \quad (14)$$

where, $u_{\text{c},s}(0)$ is the subsurface current velocity at the sea surface; β is

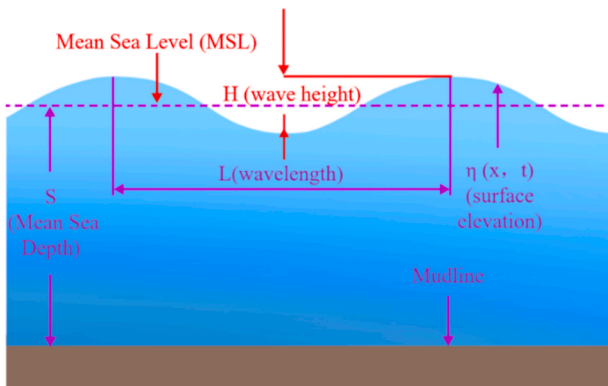


Fig. 3. Sinusoidal wave surface elevation [40].

usually taken as 1/7. Therefore, the total water particle velocity vector $u_w(z)$ is defined as the linear superposition of wave-induced velocity and current-induced velocity:

$$u_w(z) = u_{\text{wav}}(z) + u_{\text{cur}}(z) \quad (15)$$

where $u_{\text{wav}}(z)$ and $u_{\text{cur}}(z)$ are the water particle velocity vectors induced by waves and currents, respectively.

In accordance with the steady-state current assumption specified in the DNV-RP-C205 [41] standard, particle acceleration originates solely wave-induced. At time t and position (x, z) , the acceleration components of a water particle are given by:

$$\begin{aligned} a_{wx} &= \frac{\omega_0^2 H \cosh(k(z+S))}{2 \sinh(kS)} \sin(kx - \omega_0 t) \\ a_{wz} &= -\frac{\omega_0^2 H \sinh(k(z+S))}{2 \sinh(kS)} \cos(kx - \omega_0 t) \end{aligned} \quad (16)$$

where, a_{wx} and a_{wz} are acceleration components of water particles in x -axis and z -axis, respectively.

2.1.2. Parked condition of OWTs

Under the parked state of OWTs, the blades are feathered to a 90° pitch angle to reduce aerodynamic load. The support structure experiences extreme environmental from combined wind and wave excitations. These design requirements are codified in international design standards specified by IEC 61400-3-1 [35] and DNV-RP-C205 [41] for extreme conditions associated with 50-year return period events.

2.1.2.1. Extreme wind speed model. The extreme wind speed model (EWM) in IEC 61400-3-1 [35] for OWTs establishes the design load cases for structural analysis under parked conditions. The wind field is parameterized by the mean wind speed and turbulence standard deviation evaluated at hub height. For the extreme wind speed model, the 10-min average wind speeds correlating with 50-year and 1-year return periods, u_{e50} and u_{e1} , are calculated using:

$$\begin{aligned} u_{e50}(z) &= u_{\text{ref}} \left(\frac{z}{z_{\text{hub}}}\right)^{0.11} \\ u_{e1}(z) &= 0.8 u_{e50}(z) \end{aligned} \quad (17)$$

where, u_{ref} is the reference wind speed. The standard deviation of the longitudinal fluctuating wind speed component, denoted as σ_1 , is defined as follows:

$$\sigma_1 = 0.11 \bar{U}_{\text{hub}} \quad (18)$$

2.1.2.2. Severe sea state (SSS). The significant wave height represents the average height of the highest one-third of waves recorded over a 3-h observation window. The 50-year significant wave height, $H_{s,50}$, is the wave height statistically expected to occur once every 50 years. According to IEC 61400-3-1 [35], the spectral peak wave period corresponding to the 50-year return-period significant wave height, $T_{s,50}$, is calculated as:

$$T_{s,50} = 11.1 \sqrt{\frac{H_{s,50}}{g}} \quad (19)$$

where, $H_{s,50}$ denotes the 50-year significant wave height. The 50-year maximum wave height, $H_{m,50}$, and the corresponding spectral peak wave period, $T_{m,50}$, can be determined using Eq. (20):

$$\begin{aligned} H_{m,50} &= H_{s,50} \sqrt{\frac{\ln(N)}{2}} \\ T_{m,50} &= 11.1 \sqrt{\frac{H_{m,50}}{g}} \end{aligned} \quad (20)$$

where, N represents the number of load cycles within a 3-h interval (10,800 s), calculated as $N = \frac{10800}{T_{S,50}}$. Similarly, the 1-year return-period significant wave height $H_{S,1}$ and its spectral peak wave period $T_{S,1}$ can be determined using Eq. (21), respectively:

$$\begin{aligned} H_{S,1} &= 0.8H_{S,50} \\ T_{S,1} &= 11.1\sqrt{\frac{H_{S,1}}{g}} \end{aligned} \quad (21)$$

2.2. Working flow of jacket design

The internal forces of jacket members exhibit spatial variation along their longitudinal axes. To simplify the design process, the jacket is divided into layers, as shown in Fig. 1. While variable-section members optimize material efficiency, this study employs discrete-layer members with distinct cross-sections for preliminary design. The maximum internal forces at layer interfaces are adopted as the design values for members of the upper layer. Consequently, legs and braces within each layer were assigned uniform cross-sectional properties.

The cross-section design of jacket members requires internal force analysis, while hydrodynamic loads depend on member diameters, creating an inherently iterative process. Initially, hydrodynamic loads are evaluated using the Morison equation based on the initial dimensions of the jacket. Aerodynamic loads are then superimposed with hydrodynamic loads to perform structural analysis, quantify member stresses, and conduct capacity verification against codified strength criteria. Upon satisfying all design requirements, the process progresses; alternatively, a systematic enlargement of member diameter and wall thickness is initiated, with iterative recalculations continuing until code compliance is achieved.

The jacket structure comprises interdependent members. Crucially, hydrodynamic loads on upper-layer members propagate cumulatively downward, while lower-layer hydrodynamic interactions remain localized. Therefore, this study implements a sequential top-to-down design process: conducting cross-sectional design layer by layer from top to bottom. For the layer i members, the aforementioned iterative algorithm determines geometric dimensions satisfying capacity constraints prior to advancing to subsequent layers. The approach fixes design parameters for preceding $(i-1)$ layers during current layer optimization, stabilizing their hydrodynamic load. This computational decoupling restricts iterative to the active design layer, yielding substantial efficiency. The design procedure follows:

Step I: Prevailing industry standards and geometric relationships govern the determination of overall geometric parameters of the jacket. The cross-sectional dimensions of all members are subsequently initialized based on the fundamental frequency requirement.

Step II: Member cross-sectional dimensions are determined through a sequential top-to-down design approach, integrating axial compression capacity requirements via a coupled iterative process involving internal force calculation and section design. Specifically:

- II-1 Initialize loop indices $i = 0, j = 0$;
- II-2 Advance to design the next layer $i = i + 1$;
- II-3 Begin the j -th iteration for members in layer i ;
- II-4 Compute aerodynamic and hydrodynamic loads applied to JOWTs;
- II-5 Extract maximum internal forces of members within layer i and evaluate their load-bearing capacity;
- II-6 Verify member strength requirements. If satisfied, proceed; otherwise, enlarge cross-sections and return to Step II-3;
- II-7 Confirm jacket design completion. If achieved, proceed; otherwise, return to Step II-2.

Step III: Perform pile foundation design using the updated jacket dimensions.

Step IV: Validate the system's fundamental frequency and tower-top displacement.

In the preliminary design stage, member cross-sections have not yet been determined, precluding the development of a complete time-domain dynamic model. Consequently, this study employs a computationally efficient quasi-static approach. First, time-varying wind and wave loads under governing cases (Section 2.4) are calculated. Then, their extreme values are identified as the design load effects. Finally, these extreme loads are applied statically to the structure for internal force assessment and member sizing. To account conservatively for potential dynamic amplification, a Dynamic Amplification Factor (DAF) is applied to wave-induced load effects during assessment. Additionally, a conservative simultaneous peak assumption is adopted for combining wind and wave loads, thereby providing a practical and safe basis for preliminary design.

These steps involve some integral and iterative calculations. A program implementing this initial design method was developed using MATLAB. The complete code is accessible at <https://github.com/xirenqi/Design-of-jacket-for-OWTs>. The computational workflow is illustrated schematically in Fig. 4.

2.3. Initialization of geometric and cross-sectional dimensions of jacket

The calculation of hydrodynamic loads on a jacket structure requires both its global geometric dimensions and the cross-sectional properties

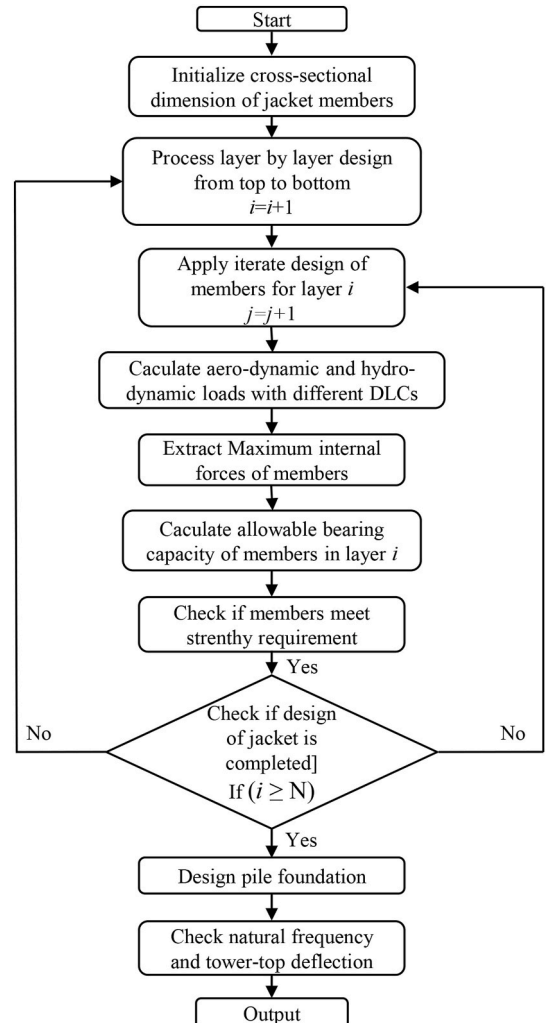


Fig. 4. Flowchart of variable cross-section jacket design.

of individual members. Typically, the global geometric parameters of the jacket are derived from industry standards and established engineering practices. In the present study, the key geometric parameters of the jacket structure used in the analysis are summarized in Table 2. A schematic of the external dimensions of the jacket structure is shown in Fig. 5. To facilitate the layered design methodology, the structural floors are sequentially indexed from top to bottom, assigning floor 1 to the uppermost level unless otherwise specified.

2.3.1. Height of jacket platform

The jacket platform height, denoted as h_j , is defined as the vertical distance from the seabed mudline to the top of the jacket. This parameter is determined to ensure the jacket's uppermost elevation remains above the waterline throughout its design lifetime. According to DNV-OS-J101 [38], h_j is calculated using Eq. (22):

$$h_j = H_{WL} + \eta_{50} + H_g = S + H_{m,50} + 0.2H_{s,50} \quad (22)$$

where, H_{WL} is the 50-year return-period high water level; η_{50} is the wave crest height; H_g is the air gap between the wave crest and the jacket top. The air gap H_g is typically set to 0.2 times the 50-year significant wave height.

2.3.2. Width of jacket top

The transition piece possesses complex geometry and is subject to loading conditions, and currently, no consensus exists within the wind energy industry regarding its optimal design methodology. The development of a simplified design framework for this component remains challenging in offshore engineering. Consequently, this study focuses exclusively on primary structural components. Consistent with established engineering practice, the tower base diameter is increased by a prescribed increment to accommodate both the transition piece installation and service platform space requirements, as expressed in Eq. (23):

$$L_{\text{bottom}} = L_{\text{top}} + \sqrt{2}h_j \cdot \tan(\alpha_v) \quad (23)$$

$$m = \left(\frac{L_{\text{bottom}}}{L_{\text{top}}} \right)^{\frac{1}{N}} L_i = m L_{i-1} = m^i L_1 h_1 = h_j \frac{m-1}{m^N-1} h_{i+1} = m L_i = m^i h_1, \text{ for } i = 1, 2, \dots, N \theta_h = \tan^{-1} \left(\frac{\sqrt{1 + \frac{\tan^2 \alpha_v}{2}}}{\frac{L_{\text{top}}}{h_1} - \frac{\sqrt{2}}{2} \tan \alpha_v} \right) \quad (24)$$

$$L_t = D_b + L_g \quad (23)$$

where, D_b is the tower-base outer diameter; L_g is the distance between the outer edges of the tower base and the jacket's top working platform.

Table 2
Primary geometric parameters of the jacket structure.

Variable and its unit	Symbol	Description
Platform height (m)	h_j	Vertical distance from the base to the top of the jacket structure
Bottom width of jacket (m)	L_b	Horizontal width at the base of the jacket
Top width of jacket (m)	L_t	Horizontal width at the top of the jacket
Angle between diagonal braces and horizontal (deg)	θ_h	Inclination angle of diagonal braces relative to the horizontal plane
Angle between main legs and vertical (deg)	α_v	Inclination angle of the jacket's main legs relative to the vertical axis
Height of layer i (m)	h_i	Vertical height of layer i of the jacket (where i is the layer number)
Bottom width of layer i (m)	L_i	Horizontal width at the base of layer i

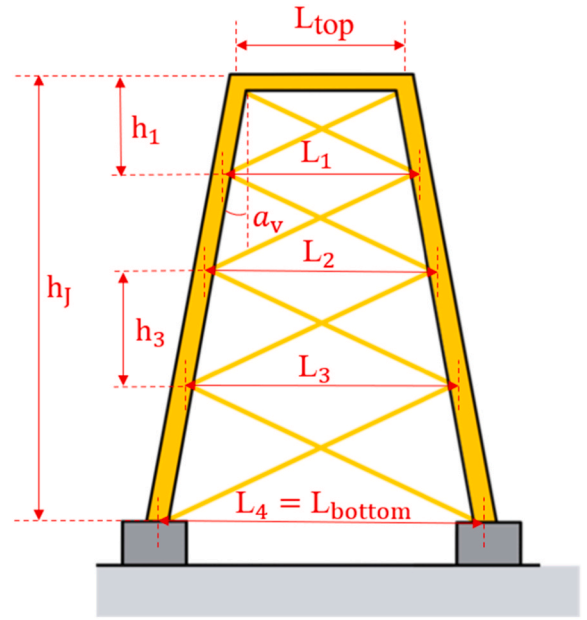


Fig. 5. Key geometric parameters of the jacket structure.

2.3.3. Dimension of members

The inclination angle of the jacket's main legs α_v is typically determined according to established oil and gas industry practices. The tangent of this angle ($\tan \alpha_v$) generally ranges from 1/8 to 1/7. Subsequently, applying geometric similarity principles [42] in conjunction with the jacket's total height, top width, and main leg inclination, the following dimensions are derived:

where, m is the width ratio between adjacent layers, defined as the lower layer width divided by the upper layer width.

2.3.4. Initialized cross-section of leg and brace

The support structures of modern large-scale OWT typically adopt a soft-stiff design philosophy. This approach aims to constrain the system's fundamental frequency between the 1P (rotor rotational frequency) and 3P frequency (blade-passing frequency) bands to avoid resonance. This strategy mitigates resonance risks by ensuring that the dominant excitation frequencies, dominated by wave, wind, and rotor dynamics, do not overlap with the structure's natural frequencies. In this study, the fundamental frequency is maintained within a specified range around the target frequency f_{fb} . During the preliminary design stage, this frequency is estimated using the expression proposed by Jalbi and Bhat-tacharya [43]:

$$f_{fb} = \frac{1}{2\pi} \sqrt{\frac{3EI_{TJ}}{(0.243m_{eq}h_{total} + M_{RNA})(h_{total})^3}} \quad (25)$$

where, EL_{TJ} is the equivalent bending stiffness of the tower-jacket system, and m_{eq} is the system's equivalent distributed mass; h_{total} is the

height of OWTs support structure; M_{RNA} is the mass of rotor-nacelle assembly. Studies indicate that the tower mass m_T dominates m_{eq} relative to the jacket mass m_J in determining the system's dynamic behavior [43]. As jacket cross-sectional dimensions remain undetermined during the preliminary design, m_{eq} is conservatively approximated as the tower's distributed mass m_T due to its dominant contribution.

By calibrating the system's fundamental frequency to the target value f_{ip} , the bending stiffness EI_{TJ} is derived from Eq. (26). Subsequently, solving Eq. (26) yields the required cross-sectional area of the jacket's main legs.

$$EI_{TJ} = EI_T \cdot \left(\frac{1}{1 + (1 + \Psi)^3 \chi - \chi} \right) \cdot \left(\frac{h_J + h_T}{h_T} \right)^3 \quad (26)$$

where, h_T is the height of the tower; EI_T is the equivalent bending stiffness of the tower; $\Psi = \frac{h_J}{h_T}$ is the height ratio of jacket and tower; $\chi = \frac{EI_T}{EI_J}$ is equivalent stiffness ratio of tower and jacket; EI_J is the equivalent bending stiffness of the jacket; the equivalent stiffness ratio of tower and jacket is given by Eq. (28):

$$EI_T = E_T I_T^{top} \cdot f(q) \quad (27)$$

$$EI_J = E_J I_J^{top} \cdot f(n) \quad (28)$$

where, E_T and E_J are the elastic moduli of the tower and jacket materials, respectively; I_T^{top} and I_J^{top} are the moments of inertia at the top cross-sections of the tower and jacket, calculated using Eqs. (29) and (30), respectively; $q = \frac{D_b}{D_t}$ is the ratio of the tower top to bottom outer diameters; D_t is tower-top outer diameter; $n = \frac{L_b}{L_t}$ is the ratio of the bottom width to the top width of the jacket; and the coefficients $f(q)$ and $f(n)$ are computed by Eqs. (31) and (32), respectively.

$$I_T^{top} = \frac{\pi}{8} D_t^3 t_T \quad (29)$$

$$I_J^{top} = A_L (L_t)^2 \quad (30)$$

$$f(q) = \frac{1}{3} \frac{2q^2(q-1)^3}{q^2(2 \ln q - 3) + 4q - 1} \quad (31)$$

$$f(n) = \frac{1}{3} \frac{n(n-1)^3}{n^2 - 2n \ln(n) - 1} \quad (32)$$

where, $t_T = \frac{m_T}{\rho_s \pi h_T D_t}$ represents the equivalent wall thickness of the tower shell; m_T is the mass of the tower; ρ_s is the density of steel; $D_T = \frac{D_t + D_b}{2}$ is the average outer diameter of tower; A_L is the cross-sectional area at the top of the jacket leg.

It should be noted that the contribution of the main legs' moment of inertia about their centroidal axes is considered negligible when calculating I_J^{top} at this design stage. According to Ref. [44], the cross-sectional area ratio between diagonal braces and main legs is initialized as 0.2:

$$\frac{A_b}{A_L} = 0.2 \quad (33)$$

where, A_b and A_L are the cross-sectional areas of the diagonal braces and main legs of the jacket structure, respectively. Similarly, in Ref. [45] recommends setting the ratio of their outer diameters to 0.4:

$$\frac{D_b}{D_L} = 0.4 \quad (34)$$

where, D_b and D_L are the outer diameters of the diagonal braces and main legs, respectively.

2.4. Loadings on jacket structure of OWT

Following the geometric initialization of jacket components, the external loads acting on the structures are determined. The OWT operates in combined atmospheric and marine environments, experiencing dynamic loading induced by wind, waves, and other environmental excitations. Although full aero-servo-hydro-elastic coupled analyses are typically required in engineering design, this study focuses on obtaining preliminary jacket design with minimal input parameters. The existing research confirms that wind and wave loads dominate the environmental loading [46]. Accordingly, this work considers three primary load categories: aerodynamic loads, hydrodynamic loads, and structural self-weight.

2.4.1. Aerodynamic loading

Given that the dominant frequency of wind loads is lower than the fundamental frequency of the wind turbine system, and to reduce computational complexity, wind loads are modeled as quasi-static excitation in this study. This approach is consistent with established practices for preliminary design of offshore wind turbine support structures [36]. The aerodynamic thrust on the rotor is calculated as:

$$T_h = \frac{1}{2} \rho_a A_R C_T (\bar{U}_{hub} + u)^2 \quad (35)$$

where, ρ_a is the air density, A_R is the rotor swept area, and C_T is the thrust coefficient. This equation is derived based on the one-dimensional quasi-steady flow assumption, and its detailed derivation can be found in Ref. [47]. The application of this quasi-steady assumption aligns with established practices for the preliminary design of offshore wind turbine support structures [36] and the analysis of OWT dynamic response [48–50]. Consequently, the structural member forces induced by aerodynamic loads are determined through static analysis in subsequent steps. This approach is justified for preliminary design stages where static analysis provides sufficient accuracy while significantly reducing computational demand.

The thrust coefficient can be obtained from wind tunnel testing, numerical simulation, or technical documentation provided for commercial turbines. In cases where such data are unavailable, empirical correlations may be applied, such as the expression proposed by Frohboese et al. [51]:

$$C_T = \frac{3.5(2\bar{U}_{hub} + 3.5)}{\bar{U}_{hub}^2} \quad (36)$$

The applicability of empirical correlations must be validated prior to use.

After computing the turbulent wind speed, the maximum wind load, $F_{wind,max}$, is evaluated using Eq. (37):

$$F_{wind,max} = \frac{1}{2} \rho_a A_R C_T (\bar{U} + u)^2 \quad (37)$$

where, ρ_a is the density of air; A_R is the sweep area of the rotor. The resultant bending-moment distribution along the jacket elevation is given by Eq. (38):

$$M_{wind,max} = F_{wind,max} (z_{hub} - z) \quad (38)$$

where, $M_{wind,max}$ is the maximum wind-induced bending moments at height z of the jacket.

2.4.2. Hydrodynamic loading

In this study, wave-induced loads are evaluated for preliminary design in a quasi-static manner. This involves calculating time-varying hydrodynamic loads based on regular waves and adopting their extreme values as design-relevant load effects. While the analysis itself is static, a dynamic amplification factor (DAF) is conservatively applied to

account for potential dynamic effects.

The Morison equation [52], a widely adopted method in offshore engineering, is used to calculate hydrodynamic loads on slender vertical cylindrical structures by superposing inertial and drag force components. The equation is expressed as:

$$f_{\text{Hydro}} = f_D + f_I = \frac{1}{2} \rho_w D_m C_D u_w |u_w| + \frac{1}{4} \rho_w \pi D_m^2 C_m a_w \quad (39)$$

where, D_m is the member diameter; ρ_w is the water density; C_D is the drag coefficient; C_m is the inertia coefficient; a_w is the water particle acceleration vector, whose components are defined in Eq. (16). According to API recommendations, C_D ranges between 0.6 and 1.0 and C_m values between 1.5 and 2.0; this study adopts the conservative values of $C_D = 1.0$ and $C_m = 2.0$ to ensure robustness in design.

As all jacket members are inclined cylindrical elements, the water particle velocity and acceleration vectors must be projected onto the normal direction of each member. The method proposed by Ref. [53] is used in the present study. For an arbitrarily oriented cylinder shown in Fig. 6, the projections of the water particle velocity vector onto its normal direction are given by Eq. (40):

$$\begin{aligned} u_{nx} &= u_x - c_x(c_x u_x + c_z u_z) \\ u_{ny} &= c_y(c_x u_x + c_z u_z) u_{nz} = u_z - c_z(c_x u_x + c_z u_z) \end{aligned} \quad (40)$$

where, u_{nx} , u_{ny} and u_{nz} are the projections of the water particle velocity vector onto the cylinder's normal direction onto the x-, y-, and z-axes, respectively; u_x and u_z represent the water particle velocity components in the global x- and z-directions. The directional coefficients c_x , c_y and c_z , relating the cylinder's orientation to the global coordinate system, are defined as:

$$\begin{aligned} c_x &= \sin \varphi \cos \theta \\ c_y &= \sin \varphi \sin \theta c_z = \cos \varphi \end{aligned} \quad (41)$$

where, φ is the inclination angle between the member and the positive z-axis; θ is the azimuthal angle between the projection of the member onto the xy-plane and the positive x-axis. The magnitude of the water particle velocity along the cylinder's normal direction, V_n is calculated as:

$$V_n = \sqrt{(u_x)^2 + (u_z)^2 - (c_x u_x + c_z u_z)^2} \quad (42)$$

The projections of the water particle acceleration vector onto the cylinder's normal direction along the x-, y-, and z-axes are expressed by Eq. (43):

$$\begin{aligned} a_{nx} &= a_x - c_x(c_x a_x + c_z a_z) \\ a_{ny} &= -c_y(c_x a_x + c_z a_z) a_{nz} = a_z - c_z(c_x a_x + c_y a_z) \end{aligned} \quad (43)$$

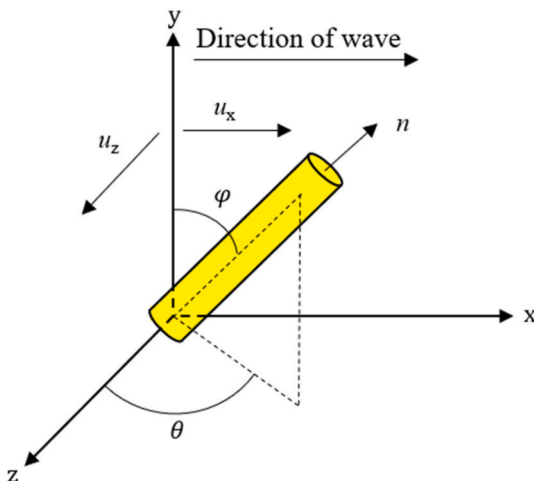


Fig. 6. Cylinder of arbitrary orientation under wave loading.

where, a_{nx} , a_{ny} and a_{nz} represent the projections of the water particle acceleration vector onto the cylinder's normal direction onto the x-, y-, and z-axes, respectively; a_x and a_z denote the water particle acceleration components in the global x- and z-axes. Subsequently, the hydrodynamic loads on the inclined cylindrical member, f_{Hydro} , are calculated using the Morison equation:

$$\begin{aligned} f_x &= \frac{1}{2} \rho_w D_m C_D V_n u_{nx} + \frac{1}{4} \rho_w \pi D_m^2 C_m a_{nx} \\ f_y &= \frac{1}{2} \rho_w D_m C_D V_n u_{ny} + \frac{1}{4} \rho_w \pi D_m^2 C_m a_{ny} \\ f_z &= \frac{1}{2} \rho_w D_m C_D V_n u_{nz} + \frac{1}{4} \rho_w \pi D_m^2 C_m a_{nz} \end{aligned} \quad (44)$$

where, f_x , f_y and f_z represent the components of the hydrodynamic load per unit length, f_{Hydro} , in the x-, y- and z-axes, respectively. At any given time t , the total hydrodynamic load on a member, F_m , is obtained by integrating the distributed load along its length:

$$\begin{aligned} F_{m,x} &= \int_L f_x dl \\ F_{m,y} &= \int_L f_y dl \\ F_{m,z} &= \int_L f_z dl \end{aligned} \quad (45)$$

where, dl is the length of the element; $F_{m,x}$, $F_{m,y}$ and $F_{m,z}$ represent the components of the hydrodynamic load, F_m , in the x-, y- and z-axes, respectively. For members near the free surface, the dynamic variation in free surface elevation must be considered. In such cases, the structure is discretized into finite elements, and the submerged length of each element is dynamically evaluated, as illustrated in Fig. 7. If an element intersects the free surface, the hydrodynamic loads are integrated only over the submerged portion.

2.4.3. Other loadings and parameters

2.4.3.1. Gravity load. The JOWT consists of four principal subsystems: rotor-nacelle assembly, tower, jacket structure, and foundation. The self-weight of these components constitutes the primary vertical loads. The gravitational load of jacket members depends on their outer diameter and wall thickness. Consequently, the structural self-weight must be recalculated following any member design updates [54,55]. For submerged members, buoyancy effects require consideration. The effective weight per unit length of a submerged, accounting for both self-weight and buoyancy, is given by:

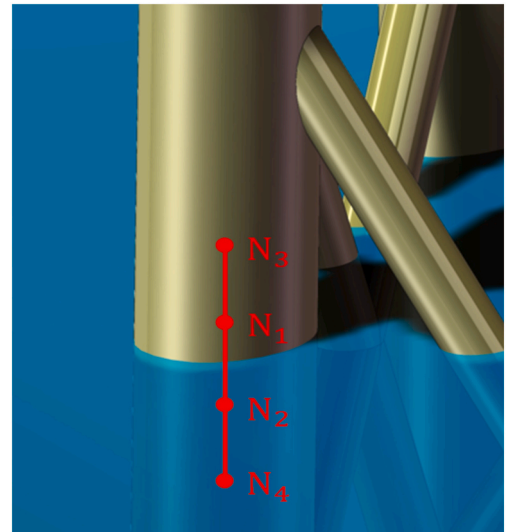


Fig. 7. Schematic of wave loading on a partially submerged member.

$$W = \frac{1}{4}\pi(D_m^2 - (D_m - 2t_c)^2)(\gamma_s - \gamma_w) \quad (46)$$

where, t_c is the thickness of cross-section; γ_s and γ_w are the unit weight of steel and seawater, respectively. Critically, wave kinematics induce dynamic variations in free surface elevation over time. Therefore, gravitational load calculations for the jacket structure must evaluate each member's position relative to the instantaneous free surface.

2.4.3.2. Dynamic magnification factor. When analyzing wave-induced loads, the dynamic amplification factor (DAF) is introduced to account for inertial and resonance effects. According to DNV-OS-J101, the DAF is formulated as:

$$DAF = \frac{1}{\sqrt{\left(1 - \left(\frac{f}{f_0}\right)^2\right)^2 + \left(2\xi\frac{f}{f_0}\right)^2}} \quad (47)$$

where, f is the load frequency, f_0 is the fundamental frequency of the system, and ξ is the critical damping ratio.

2.4.3.3. Damping ratio of the system. The damping system of OWTs incorporates structural, foundation, aerodynamic, and hydrodynamic damping components. The inclusion of these damping mechanisms, either partially or fully, depends on the operational condition of OWTs. For turbines equipped with supplemental damping devices, e.g., tuned mass damper, their damping contribution must be explicitly integrated into the system.

The structural damping of modern OWTs primarily originates from material energy dissipation. For instance, steel towers exhibit material damping ratios of 0.2%–0.3% according to industry standards [38,56]. The foundation damping consists of soil hysteretic damping and radiation damping, with their magnitudes governed by soil type, stratigraphy, and load characteristics. Carswell et al. [57] and standards [56] suggest neglecting radiation damping for excitation frequencies below 1 Hz. Building on this premise, Tarp-Johansen et al. [58] modeled foundation damping using viscous damping and Rayleigh damping formulations, proposing damping ratios of 0.5%–0.8% for OWTs foundations.

The OWTs in the running condition experience aerodynamic damping, which results from a force component proportional to and opposing the blade vibration velocity. Under the assumption of a rigid rotor and using blade element momentum (BEM) theory, Xi et al. [59] developed analytical models for fore-aft (FA) and side-to-side (SS) aerodynamic damping under uniform wind normal to the rotor plane. Their models, validated via the FAST software, indicated significant differences between FA and SS aerodynamic damping ratios during operation. For the a 5 MW and 10 MW OWTs, they recommended damping ratios of 5% (fore-aft) and 1% (side-to-side) [60]. The hydrodynamic damping primarily arises from viscous drag forces and wave radiation effects. DNV-OS-J101 [38] recommends a wave radiation damping ratio of 0.15% for such applications.

Consequently, the system damping ratio of OWTs is influenced by operational conditions and site-specific geotechnical parameters. Based on empirical and numerical studies [61], this study adopts system damping ratios of 5% for running conditions and 1% for parked conditions, consistent with established practice for offshore wind turbine design.

2.5. Cross-section design of jacket member

After determining the aerodynamic and hydrodynamic loads on the structure, a simplified analysis of member internal forces is performed. The global load distribution is illustrated in Fig. 8. It is important to note that both aerodynamic and hydrodynamic loads act as distributed loads; their representation as concentrated forces in the figure is solely for

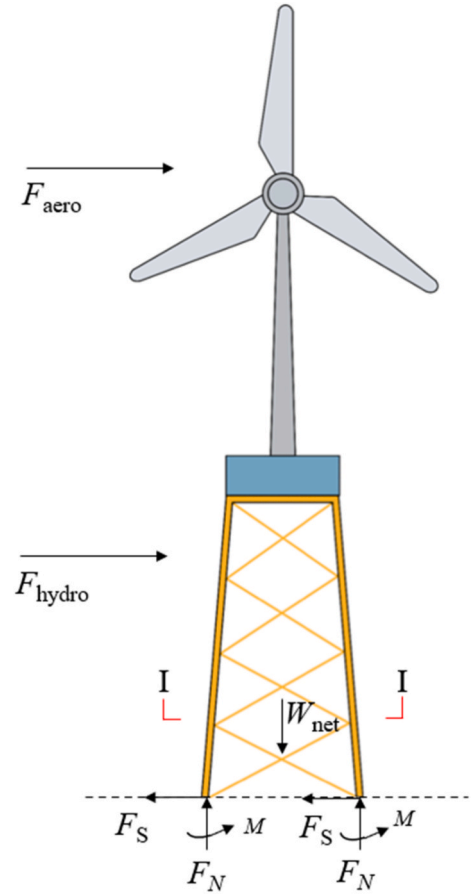


Fig. 8. Load analysis of jacket structure.

visualization only. To account for wind-wave misalignment and nacelle yaw effects, load combinations under directional scenarios are assessed. This study considers four primary load cases: (a) co-directional wind and wave loads; (b) wind and wave loads with a directional discrepancy of 90°; (c) wind and wave loads with a directional offset of 45°; (d) wind and wave loads both acting simultaneously at 45° to the structure.

To simplify the internal force analysis of main legs, diagonal braces, and pile tops, all structural members are idealized as pin-connected truss elements with flexural rigidity neglected. External loads are applied exclusively at nodal points, ensuring compliance with the truss assumption and simplifying the analytical model. Consequently, the structural members and piles are slender components that primarily resist axial tension and compression under combined environmental and gravitational loading.

2.5.1. Internal force of leg

The internal forces in the main legs are calculated based on the loading schematic in Fig. 9, considering the cross-section located at the base of layer i (Section I-I in Fig. 8). In Fig. 9, F_x and F_y are the components of the resultant force acting on the jacket cross-section along the global x- and y-axes, respectively; M_x and M_y are the components of the resultant moment acting on the jacket cross-section about the global x- and y-axes, respectively; F_{aero} and F_{hydro} are the resultant aerodynamic and hydrodynamic load, respectively; M_{aero} and M_{hydro} are the corresponding resultant aerodynamic and hydrodynamic moment, respectively.

The resultant aerodynamic and hydrodynamic load are simplified to act through the centroid of this section, with the corresponding resultant moments. Decomposing these loads into components in the x- and y-

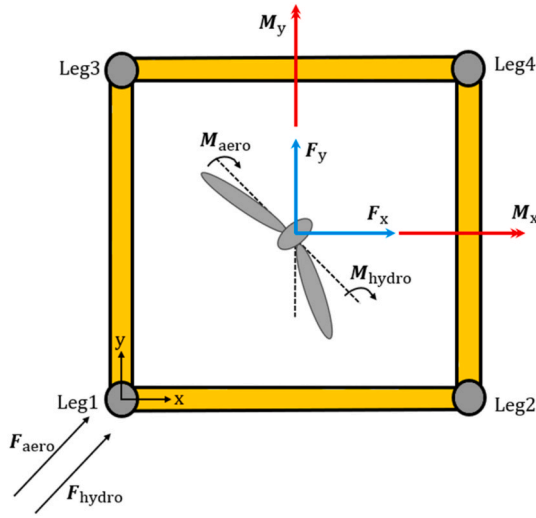


Fig. 9. Schematic diagram for internal force calculation in main legs.

directions, the resultant forces and moments at the section are formulated as:

$$\begin{aligned} F_x &= F_{aero} \cdot \cos(\beta_1) + F_{hydro} \cdot \cos(\beta_2) \\ F_y &= F_{aero} \cdot \sin(\beta_1) + F_{hydro} \cdot \sin(\beta_2) \\ M_x &= M_{aero} \cdot \sin(\beta_1) + M_{hydro} \cdot \sin(\beta_2) \\ M_y &= M_{aero} \cdot \cos(\beta_1) + M_{hydro} \cdot \cos(\beta_2) \end{aligned} \quad (48)$$

where, β_1 and β_2 are the dominant wind direction and the wave-current propagation direction, respectively.

Under gravitational loading, the axial forces in legs are derived as:

$$F_{N,g} = \frac{(W_{RNA} + W_T + W_J + W_{TP})}{4} \quad (49)$$

where, $F_{N,g}$ is the axial forces in the main legs induced by gravitational loads; W_{RNA} , W_T , W_J , and W_{TP} represent the gravitational loads of the rotor-nacelle assembly, tower, jacket structure, and transition piece, respectively. These loads are cumulatively transmitted through the jacket's main legs, maintaining vertical equilibrium under self-weight conditions.

Assuming that the shear forces are uniformly distributed among all main legs, the shear force carried by each leg can be expressed as:

$$\begin{aligned} F_{sx} &= \frac{F_x}{4} \\ F_{sy} &= \frac{F_y}{4} \end{aligned} \quad (50)$$

where, F_{sx} and F_{sy} are the shear forces along the x- and y-directions, respectively, acting on the leg cross-section. Under combined wave and gravitational loading, the axial forces govern the structural integrity. From moment equilibrium at section I-I (Fig. 8), the axial forces of legs 1 and 4 (Fig. 9) are:

$$\begin{aligned} F_{N,i1} &= \frac{M_x + M_y}{2L_i} - F_{N,g} \\ F_{N,i4} &= -\frac{M_x + M_y}{2L_i} - F_{N,g} \end{aligned} \quad (51)$$

where, $F_{N,i1}$ and $F_{N,i4}$ are the axial forces of legs 1 and 4 for the layer i , respectively. Critically, the axial and shear forces at the pile-top equal

the corresponding axial and shear forces in the legs at the base of the jacket structure.

2.5.2. Internal forces of brace

When calculating the internal forces of braces, the jacket structure is simplified as planar truss systems. For three-dimensional jackets, each structural face is treated as an independent planar truss. By decomposing the entire structure into multiple planar trusses, the computational process is simplified [62], enabling independent analysis of internal forces within each planar truss. This methodology significantly improves computational efficiency for complex structures, as sequential analysis and design of individual planar trusses can be executed without simultaneous treatment of the structural system. The analytical framework for diagonal brace internal forces is shown in Fig. 10.

Considering the equilibrium of horizontal forces, the following relationship is derived:

$$2F_{sx} = 2F_{Nb} \cos \theta_h \Rightarrow F_{Nb} = \frac{F_{sx}}{\cos \theta_h} \quad (52)$$

where, F_{Nb} is the axial force of the brace. Under the excitation of shear force F_{sx} , the paired braces in Fig. 10 develop opposing internal forces: one member sustains tension while the counterpart undergoes compression, depending on the load direction. Since these braces are highly flexible members, only their compressive capacity is evaluated in the subsequent analysis.

2.5.3. Cross-section design of member

As the legs and braces of jacket are typically slender members with high geometric slenderness, their cross-sectional design must prioritize buckling resistance under compressive-dominated conditions. The method p recommended by API specifications [63] is used to calculate the allowable compressive stress of tubular members, as shown in Eq. (53):

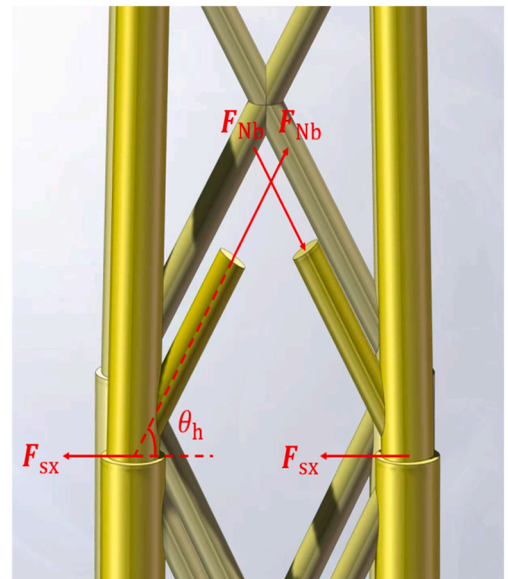


Fig. 10. Schematic diagram for internal force calculation in diagonal braces.

$$\sigma_{\text{allow}} = \begin{cases} \left(1 - \left(\frac{\mu L}{\sqrt{2} r C_c}\right)^2\right) f_y, & \frac{\mu L}{r} < C_c \\ \frac{5}{3} + \frac{3\mu L}{8r C_c} - \frac{1}{8} \left(\frac{\mu L}{C_c r}\right)^3, & \frac{\mu L}{r} < C_c \\ \frac{12\pi^2 E}{23 \left(\frac{\mu L}{r}\right)^2}, & \frac{\mu L}{r} \geq C_c \end{cases} \quad (53)$$

where, μ is the recommended effective length factor, and it is setting as 0.8 in the present study; L is the longest unsupported length of the lowest segment; r is the radius of gyration of the initial support cross-section calculated previously; $\frac{\mu L}{r}$ is the slenderness ratio; f_y is the yield stress; E is the modulus of elasticity; C_c is the critical slenderness, defined as $C_c = \sqrt{\frac{2E}{\pi^2 f_y}}$.

Subsequently, the axial capacity of both legs and braces must be systematically verify to ensure structural integrity. If the calculated axial load is less than the allowable axial load, subsequent analytical procedures may be executed. Otherwise, structural reinforcement through cross-sectional dimension augmentation must be implemented, followed by iterative recalibration until compliance with prescribed limits is achieved.

2.6. Capacity of pile foundation

The DNVGL-RP-C212 standard recommends several pile capacity calculation methods for different soil types [64], including: ICP method [65], NGI method [66], and API method [63]. Due to its simplicity and practical applicability, the API method is adopted in this study for the design of jacket pile foundations. This section provides a brief overview of the calculation methods for pile capacity in both non-cohesive and cohesive soils.

2.6.1. Cohesionless soils

For cohesionless soils, the unit shaft friction, f_p , expressed in stress units, can be calculated using Eq. (54):

$$f_p = K \sigma'_v \tan \delta \quad (54)$$

where, K is the lateral earth pressure coefficient; σ'_v denotes the effective vertical stress of the soil layer; δ is the soil-pile interface friction angle.

The ultimate pile-tip axial bearing capacity per unit area, q_u , is given by Eq. (55):

$$q_u = N_q \sigma'_v \quad (55)$$

where, N_q is Bearing capacity factor.

2.6.2. Cohesive soils

For cohesive soils, API recommends the " α " method, in which the unit shaft friction, f_p , in stress units, is calculated using Eq. (56):

$$f_p = \alpha c_u \quad (56)$$

where, c_u is the undrained shear strength of the soil; α is the dimensionless adhesion factor, which can be calculated using Eq. (57):

$$\alpha = 0.5 \varphi^{-0.5}, \varphi < 1 \\ \alpha = 0.5 \varphi^{-0.25}, \varphi > 1 \quad (57)$$

where, $\varphi = \frac{c_u}{\sigma'_v}$

The ultimate pile-tip axial bearing capacity per unit area, q_u , is given by Eq. (58):

$$q_u = 9c_u \quad (58)$$

2.7. Estimation of the system's natural frequency

2.7.1. Obtain the jacket stiffness

After finalizing the dimension of jacket members and pile foundations, the natural frequency must be verified using the updated jacket geometry. The derivation of the equivalent jacket stiffness is predicated on two key assumptions: (1) the braces offer sufficient shear stiffness to constrain the jacket frame, ensuring bending-dominated global deformation; while (2) the cross-section of the legs remains plane during bending, consistent with the Bernoulli-Euler beam hypothesis. Consequently, the legs of jacket are modeled as Euler-Bernoulli cantilever beams [43]. Then, the jacket is further simplified to an equivalent uniform cantilever beam through tip displacement equivalence. The simplified models of the jacket are shown in Fig. 11, where P is a concentrated load applied at the free end.

Applying the parallel axis theorem while neglecting local centroidal inertia, the moment of inertia for layer i cross-section, I_i , is given by:

$$I_i = A_{ci} \left(\frac{d_i}{2}\right)^2 \times 4 = A_{ci} d_i^2 \quad (59)$$

where, A_{ci} is the cross-sectional area of the legs in layer i of the jacket, and d_i is the horizontal distance between the axes of the two legs. The governing equation for the stepped cantilever beam with variable cross is expressed by:

$$\frac{d^2 w}{dz^2} = \frac{P(h_j - z)}{E_j A_{ci} \left(\frac{L_i - L_b}{h_j} z + L_b\right)^2} \quad (60)$$

where, w is the deflection of the beam cross-section along the horizontal

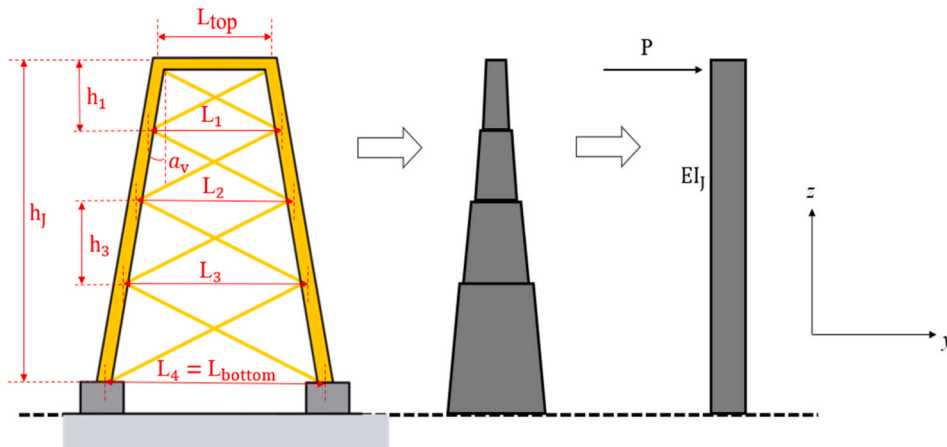


Fig. 11. Equivalent beam model of jacket structure.

direction. The boundary conditions are defined as follows:

$$\begin{aligned} w'_4|_{z=0} &= 0 \\ w_4|_{z=0} &= 0 \end{aligned} \quad (61)$$

For jacket with tapered cross-section, the displacement and slope continuity at adjacent section must be satisfied, i.e.:

$$\begin{aligned} w_4|_{z=h_4} &= w_3|_{z=h_4} \\ w'_4|_{z=h_4} &= w'_3|_{z=h_4} \\ w_3|_{z=h_4+h_3} &= w_2|_{z=h_4+h_3} \\ w'_3|_{z=h_4+h_3} &= w'_2|_{z=h_4+h_3} \\ w_2|_{z=h_4+h_3+h_2} &= w_1|_{z=h_4+h_3+h_2} \\ w'_2|_{z=h_4+h_3+h_2} &= w'_1|_{z=h_4+h_3+h_2} \end{aligned} \quad (62)$$

where, w_i and w'_i are the lateral deflection function and rotation angle function, respectively, of layer i of the jacket.

Introducing the dimensionless parameter $\lambda = \frac{L_t - L_b}{h_j}$, the integrated solutions for layer i become:

$$\begin{aligned} \frac{dw}{dz} &= \frac{P}{E_j A_{ci} \lambda^2} \left(-\frac{L_t}{\lambda z + L_b} - \ln|\lambda z + L_b| \right) + A_i \\ w(z) &= \frac{P}{E_j A_{ci} \lambda^3} \left(-(L_t + L_b + \lambda z) \ln|\lambda z + L_b| + \lambda z + L_b \right) + A_i z + B_i \end{aligned} \quad (63)$$

For Layer 4, integration constants are uniquely determined by the clamped-end boundary conditions shown in Eq. (61):

$$\begin{aligned} A_4 &= \frac{P}{E_j A_{c4} \lambda^2} \left(\frac{L_t}{L_b} + \ln|L_b| \right) \\ B_4 &= \frac{P}{E_j A_{c4} \lambda^3} \left((L_t + L_b) \ln|L_b| - L_b \right) \end{aligned} \quad (64)$$

Substituting $z = h_j$ into Eq. (63) yields tip deflection $w_1|_{z=h_j}$:

$$w_1|_{z=h_j} = \frac{P(\lambda h_j + L_b - (L_t + L_b + \lambda h_j) \ln|\lambda h_j + L_b|)}{E_j A_{c1} \lambda^3} + A_1 h_j + B_1 \quad (65)$$

For comparison, the tip deflection for the uniform cantilever beam under tip load, w_{h_j} , is:

$$w_{h_j} = \frac{PL^3}{3EI_{eq}} \quad (66)$$

Equating $w_1|_{z=h_j} = w_{h_j}$ provides:

$$\frac{PL^3}{3EI_{eq}} = \frac{P(\lambda h_j + L_b - (L_t + L_b + \lambda h_j) \ln|\lambda h_j + L_b|)}{E_j A_{c1} \lambda^3} + A_1 h_j + B_1 \quad (67)$$

Through systematic derivation, the equivalent flexural stiffness of the jacket emerges as:

$$EI_{eq} = \frac{-h_j^3}{3 \cdot (h_3 + h_4) \cdot (T_1 + T_2 + T_3 + T_4 + T_5 + T_6 + T_7 + T_8 + T_9)} \quad (68)$$

where, the mathematical expressions for T_i ($i = 1, \dots, 9$) are too lengthy; therefore, they are presented collectively in Appendix A.

Setting $EL_j = EI_{eq}$ and substituting it into Eq. (26) yields the jacket-tower equivalent stiffness, EL_{T-J} . Substituting EL_{T-J} into Eq. (25) then gives the fixed-base natural frequency of the jacket-supported offshore wind turbine f_{fb} , neglecting SSI effects. The equivalent mass required m_{eq} , in this calculation is determined using Eq. (69).

$$m_{eq} = \frac{\int m(z) \phi_1^2 dz}{\int \phi_1^2 dz} = \frac{\sum_{i=1}^n m_i \int_{z(i-1)}^{z(i)} \phi_1^2 dz}{\int_0^{z(i)} \phi_1^2 dz} = \frac{\bar{m}_j \int_0^{h_j} \phi_1^2 dz + \bar{m}_T \int_{h_j}^{h_j+h_T} \phi_1^2 dz}{\int_0^{h_j+h_T} \phi_1^2 dz} \quad (69)$$

where, $\bar{m}_j$ and $\bar{m}_T$ are the mass per unit length of the jacket and the tower, respectively, defined as the mass of each structure divided by its height; ϕ_1 is the mode function of cantilever beam.

2.7.2. Calculate the natural frequency including SSI

In structural design of the JOWT, its natural frequency must be

sufficiently separated from the dominant excitation frequencies to avoid resonance. When accounting for SSI, the system is modeled as a uniform cross-section beam supported by two vertical springs, as depicted in Fig. 12. The equivalent stiffness k_R is determined by using the energy method, as presented in Eq. (70).

$$k_R = k_V L_{bottom}^2 \left(\frac{\vartheta}{1 + \vartheta} \right) \quad (70)$$

where, $k_V = 2\pi L_{pile} G_s / \zeta$ is vertical stiffness of the foundation, G_s is the average shear modulus of soil between the top and bottom of the pile, and parameter ζ is set as 4 according to Ref. [67]; ϑ is foundation stiffness heterogeneity factor. Evidently, k_R depends on the vertical

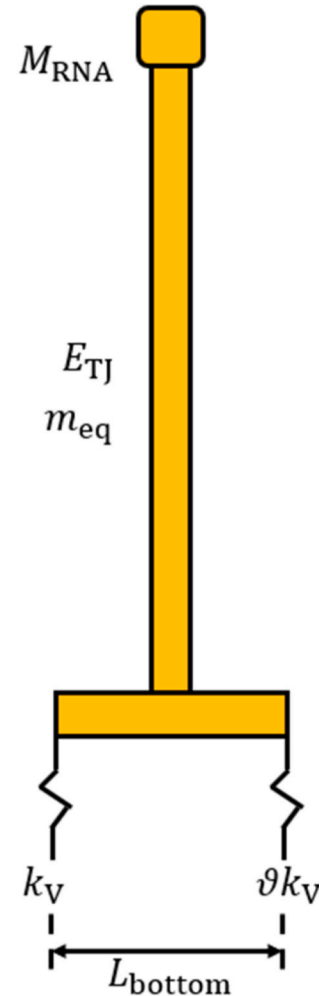


Fig. 12. Uniform beam with soil-structure interaction.

foundation stiffness and the spacing between foundation elements. The global natural frequency, f_0 , is given by the product of the fixed-base frequency f_{fb} and the flexibility coefficient C_J :

$$f_0 = C_J f_{fb} = \frac{C_J}{2\pi} \sqrt{\frac{3EI_{TJ}}{(0.243m_{eq}h_{total} + M_{RNA})h_{total}^3}} \quad (71)$$

where, $C_J = \sqrt{\frac{\tau}{\tau+3}}$ and $\tau = \frac{k_R h_{total}}{EI_{TJ}}$. The coefficient C_J quantifies the stiffness reduction attributable to foundation flexibility and is a function of the foundation's equivalent stiffness.

2.7.3. Calculate tower-top displacement considering SSI

Based on the classical Euler-Bernoulli beam theory, the tower-top deflection of can be decomposed into contributions from aerodynamic load, δ_{aero} , and hydrodynamic load, δ_{hydro} [68]. The tower-top deflection induced by wave can be calculated using Eq. (72), with the equivalent bending stiffness of the jacket.

$$\delta_{hydro} = \frac{F_{hydro}(h_{total} - a)^2}{k_R} \cdot \frac{h_{total}}{h_{total} - a} + \frac{F_{hydro}}{EI_J} \left[\frac{(h_{total}^3 - a^3)}{3} - \frac{(h_{total}^2 - a^2)a}{2} \right] \quad (72)$$

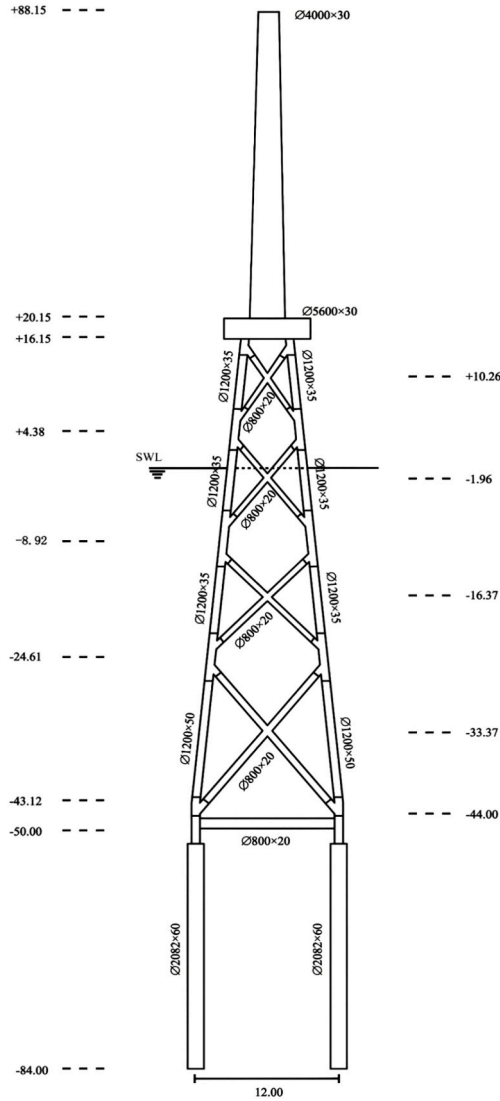


Fig. 13. Geometric dimensions of the jacket structure in study case (All dimensions of structural members are in millimeters (mm); heights and depths are specified in meters (m) unless otherwise stated).

where, h_{total} is the total height of the support structure; a is the vertical distance from F_{hydro} to the tower top. The tower-top deflection induced by wind is calculated using Eq. (73), with the equivalent bending stiffness of tower and jacket.

$$\delta_{aero} = \frac{F_{aero}h_{total}^2}{k_R} + \frac{F_{aero}h_{total}^3}{3EI_{T-J}} \quad (73)$$

2.7.4. Validation of the natural frequency calculation formula

As illustrated in Fig. 12, the 5 MW JOWT from Ref. [69] has a rotor-nacelle assembly (RNA) mass of 350 tons. To assess the reliability of the proposed fundamental frequency formulation, the structure depicted in Fig. 13 is analyzed. The key computational procedures and results are summarized below.

First, the equivalent stiffness of the support structure was determined. The Young's modulus is taken as 210 GPa, representing typical steel. Based on Eq. (28), the equivalent bending stiffness of the jacket, EI_J , is calculated as 4.372×10^{12} N·m². From the dimensions in Fig. 13, the equivalent outer diameter D_T and wall thickness t_T of the tower are 4.8 m and 0.0308 m, respectively. The moment of inertia at the tower-top section I_T^{top} is calculated to be 0.7743 m⁴. With a base-to-top cross-sectional diameter ratio $q = 1.4$, the equivalent stiffness of the tower, EI_T , is derived using Eq. (27), i.e.:

$$EI_T = E_T I_T^{top} \cdot f(q) = 2.1 \times 10^{11} \times 0.7743 \times 2.1459 = 3.489 \times 10^{11} \text{ N·m}^2$$

The dimensionless parameters χ and ψ are found to be 0.0798 and 0.8716, respectively. The equivalent bending stiffness of the jacket-tower system, EI_{TJ} , is obtained using Eq. (26):

$$EI_{TJ} = 3.489 \times 10^{11} \times \left(\frac{1}{1 + (1 + 0.8716)^3 \times 0.0798 - 0.0798} \right) \cdot \left(\frac{59.27 + 68}{68} \right)^3 = 1.5849 \times 10^{12} \text{ N·m}^2$$

Based on the masses of the tower and jacket, along with the cantilever mode shape function, the equivalent mass of the support structure, m_{eq} , is determined as 3.77 t/m. Subsequently, the fixed-base fundamental frequency f_{fb} of the JOWT system is calculated as 0.354 Hz using Eq. (25). Accounting for foundation flexibility, the equivalent foundation stiffness k_R is determined as:

$$k_R = k_v L_{bottom}^2 \left(\frac{\theta}{1 + \theta} \right) = 1.6 \times 12^2 \times \left(\frac{1}{1 + 1} \right) = 1.15 \times 10^{11} \text{ N·m}$$

Finally, substituting the parameters $\tau = 9.26$ and $C_J = 0.869$ into Eq. (71), the fundamental frequency of the system, f_0 , is determined as 0.3077 Hz. This result exhibits a 2.5% relative deviation from the 0.3 Hz natural frequency obtained through GH-Bladed software simulations in Ref. [69]. The proposed methodology demonstrates sufficient accuracy for engineering applications and is suitable for preliminary natural frequency analysis of wind turbine systems with tapered-section jacket-type supporting structures.

Table 3
Design parameters summary of 5 MW wind turbine.

Tower	Value
Tower Height, h_T (m)	70
Top Diameter, D_{top} (m)	4
Bottom Diameter, D_{bottom} (m)	5.6
Equivalent Tower Wall Thickness, t_T (m)	0.0315
Total Tower Mass, m_T (kg)	261100
Mass of RNA, M_{RNA} (kg)	350000
Site Water Depth, S (m)	50
50-Year characteristic Wave Height, $H_{s,50}$ (m)	9.4 m
50-year extreme wave height, $H_{m,50}$ (m)	17.2 m
The extreme wind speed with a 50-year return period, $U_{e,50}$ (m/s)	42.73
The extreme wind speed with a 1-year return period, $U_{e,1}$ (m/s)	34.18

3. Step-by-step solved example

To evaluate the capability of proposed methodology for generating rational preliminary designs of OWT jacket support structures, this section utilized research outcomes from European Upwind [70] and INNWIND projects [71] as benchmarks. The preliminary designs of the jacket for NREL 5 MW and DTU 10 MW OWT are developed following the approach outlined in Section 2. These benchmark projects adhere to the constraints specified in current standards (DNV-ST-0126 [36] and IEC 61400-3-1 [35]) regarding system natural frequencies, ultimate bearing capacity, and fatigue life.

3.1. Preliminary design of jacket support structure for 5 MW OWT

3.1.1. Determine the overall geometric dimensions of jacket

The key design parameters for the jacket of the NREL 5 MW OWT are summarized in Table 3, with detailed technical specifications available in Ref. [72]. Based on these parameters, the height of the jacket, h_J , is determined using Eq. (22):

$$h_J = H_{WL} + \eta_{50} + H_g = S + H_{m,50} + 0.2H_{s,50} \approx 67 \text{ m}$$

Based on the tower base diameter and transition piece dimensions, the platform top width L_t is set to 8 m. The initial main leg inclination angle, α_v , of the jacket structure is set as 2.42° based on the jacket height. Subsequently, the width ratio between adjacent layers, m , and vertical height of layer 1 of the jacket, h_1 , are calculated as:

$$m = \left(\frac{L_{\text{bottom}}}{L_{\text{top}}} \right)^{\frac{1}{N}} = \left(\frac{L_{\text{top}} + \sqrt{2} \tan \alpha_v h_J}{L_{\text{top}}} \right)^{\frac{1}{N}} = \left(\frac{8 + \sqrt{2} \tan 2.42 (67)}{8} \right)^{\frac{1}{4}} = 1.1068$$

$$h_1 = h_J \frac{m-1}{m^N-1} = 67 \times \frac{1.1068-1}{1.1068^4-1} = 14.29 \text{ m}$$

The inclination angle of the brace, θ_h , is calculated using Eq. (24). Then, the elevation and widths of each layer are derived from Eq. (24) and are presented in Table 4.

$$\theta_h = \tan^{-1} \frac{\sqrt{1 + \tan^2 \alpha_v}}{\frac{L_{\text{top}}}{h_1} - \frac{\sqrt{2}}{2} \tan \alpha_v} = \tan^{-1} \frac{\sqrt{1 + \tan^2 2.42}}{\frac{8}{14.2932} - \frac{\sqrt{2}}{2} \tan 2.42} = 62.1^\circ$$

3.1.2. Determine the initial dimensions of jacket members

The target natural frequency is set to lie between the 1P rotational and 3P frequency bands of the wind turbine. To avoid resonance, the upper bound of the system's fundamental frequency is set to 0.9 times the minimum 3P frequency, and the lower bound is set to 1.1 times the maximum 1P frequency. Therefore, the target frequency, f_{tp} , is defined as the midpoint of this range.

$$f_{3P,\min} = \frac{3 \times n_{\min} \times 2 \times \pi \times 0.9}{60 \times 2 \times \pi} = 0.3105 \text{ Hz}$$

$$f_{1P,\max} = \frac{n_{\max} \times 2 \times \pi \times 1.1}{60 \times 2 \times \pi} = 0.2218 \text{ Hz}$$

$$f_{tb} = \frac{f_{3P,\min} + f_{1P,\max}}{2} = \frac{0.3105 + 0.2218}{2} = 0.2662 \text{ Hz}$$

The equivalent stiffness of the jacket-tower system is calculated using Eq. (26):

Table 4
Summary of external dimensions for jacket of 5 MW OWT.

Layer number	Height (m)	Width (m)
1	14.29	8.85
2	15.82	9.80
3	17.51	10.85
4	19.38	12.01

$$EI_{T-J} = 1.1367 \times 10^{12} \text{ N}\cdot\text{m}^2$$

According to the average tower thickness, t_T , summarized in Table 3, the equivalent stiffness of the tower, EI_T , is calculated using Eq. (27):

$$I_T^{\text{top}} = \frac{\pi}{8} D_{\text{top}}^3 t_T = \frac{\pi}{8} \times 4^3 \times 0.0315 = 0.7919 \text{ m}^4$$

$$q = \frac{D_b}{D_t} = \frac{5.6}{4} = 1.4$$

$$f(q) = \frac{1}{3} \frac{2q^2(q-1)^3}{q^2(2 \ln q - 3) + 4q - 1} = 2.1459$$

$$EI_T = E_T I_T^{\text{top}} \cdot f(q) = 3.40 \times 10^{11} \text{ N}\cdot\text{m}^2$$

Subsequently, the initial cross-sectional area of the jacket leg, A_{c0} , necessary to satisfy the frequency criterion is derived from Eqs. (25)–(32) as follows:

$$\psi = \frac{h_J}{h_T} = \frac{67}{70} = 0.96$$

$$1.1367 \times 10^{12} = 3.40 \times 10^{11} \cdot \left(\frac{1}{1 + (1 + 0.96)^3 \chi - \chi} \right) \cdot \left(\frac{67 + 70}{70} \right)^3$$

Solving this equation yields:

$$\chi = \frac{EI_T}{EI_J} = 0.1911 \Rightarrow EI_J = 1.78 \times 10^{12} \text{ N}\cdot\text{m}^2$$

$$EI_J = E_J I_J^{\text{top}} \cdot f(m)$$

$$m = \frac{L_{\text{bottom}}}{L_{\text{top}}} = \frac{12}{8} = 1.5$$

$$f(m) = \frac{1}{3} \frac{m(m-1)^3}{m^2 - 2m \ln(m) - 1} = \frac{1.5 \times (1.5-1)^3}{1.5^2 - 2 \times 1.5 \times \ln(1.5) - 1} = 1.86$$

$$1.78 \times 10^{12} \text{ N}\cdot\text{m}^2 = E_J \times I_J^{\text{top}} \times 1.86 \Rightarrow I_J^{\text{top}} = 4.78 \text{ m}^4$$

$$I_J^{\text{top}} = A_{c0} (L_t)^2 = A_{c0} = 0.075 \text{ m}^2$$

According to Eq. (34), the brace-to-leg diameter ratio is selected as:

$$D_L = 0.619 \text{ m}$$

$$0.4 = \frac{D_b}{D_L} \Rightarrow D_b = 0.248 \text{ m}$$

3.1.3. Determine the cross-sections of legs and braces

With the preliminary jacket dimensions established, the aerodynamic and hydrodynamic loads acting on the OWT were determined. The load cases from Subsection 2.1 and the load calculation procedure from Subsection 2.4 are applied in the following analysis.

3.1.3.1. Aerodynamic load. The thrust coefficient is calculated as:

$$C_T = \frac{3.5(2U_R + 3.5)}{U_R^2} = \frac{3.5 \times (2 \times 11.4 + 3.5)}{11.4^2} = 0.71$$

To simplify the computational analysis, both the hub-height wind speed ($\bar{U}_{\text{hub}}$) and the average wind speed ($\bar{U}_{\text{avg}}$) are set to the rated wind speed, U_R . The standard deviation of turbulence wind speed for the ETM, $\sigma_{U,\text{ETM}}$, is calculated as:

$$\sigma_{U,\text{ETM}} = 2 \times 0.18 \times \left[0.072 \left(\frac{11.4}{2} + 3 \right) \left(\frac{11.4}{2} - 4 \right) + 10 \right] = 4.0$$

$$\sigma_{U,ETM,f>1p} = \sqrt{\int_{f_{1p,max}}^{\infty} S_{uu}(f) df} = \sigma_{U,ETM} \sqrt{\frac{1}{\left(\frac{6L_E f}{U_R} f_{1p,max} + 1\right)^2}} = 4$$

$$\times \sqrt{\frac{1}{\left(\frac{6 \times 340.2}{11.4} \times 0.2218 + 1\right)^2}} = 1.2$$

Assuming turbulent velocity fluctuations follow a Gaussian distribution at the 95% confidence level, the characteristic extreme turbulence velocity, u_{ETM} , is calculated using:

$$u_{ETM} = 2 \cdot \sigma_{U,ETM,f>1p} = 2 \times 1.2 = 2.4 \text{ m/s}$$

For the ETM design load case, the aerodynamic load acting on the wind turbine, F_{ETM} , is calculated using:

$$F_{ETM} = \frac{1}{2} \rho_a A_R C_T (U_R + u)^2 = \frac{1}{2} \times 1.225 \times 12469 \times 0.7083$$

$$\times (11.4 + 2.4)^2 = 1.03 \text{ MN}$$

This yielded the maximum moment acting on the bottom section of layer 1 of the jacket, M_{ETM} :

$$M_{ETM} = F_{ETM} \times (h_T + h_1) = 86.8 \text{ MN}\cdot\text{m}$$

Following a similar methodology, the aerodynamic loads (F_{EOG}) and corresponding moment (M_{EOG}) under the EOG condition are assessed with:

$$F_{EOG} = 1.74 \text{ MN}$$

$$M_{EOG} = 146.5 \text{ MN}\cdot\text{m}$$

The aerodynamic loads obtained from the 1-year ($F_{EWM,1}$) and 50-year extreme ($F_{EWM,50}$) wind speed model are:

$$F_{EWM,1} = \frac{1}{2} \rho_a A_R C_T U_{e,1}^2 = \frac{1}{2} \times 1.225 \times 12469 \times 0.06 \times 56^2 = 1.44 \text{ MN}$$

$$F_{EWM,50} = \frac{1}{2} \rho_a A_R C_T U_{e,50}^2 = \frac{1}{2} \times 1.225 \times 12469 \times 0.06 \times 70^2 = 2.25 \text{ MN}$$

3.1.3.2. Hydrodynamic load. The extreme hydrodynamic loads are determined through a cascading statistical analysis of wave heights:

$$H_{S,1} = 0.8 \times 8.24 = 6.59 \text{ m}$$

$$T_{S,1} = 11.1 \sqrt{\frac{H_{S,1}}{g}} = 11.1 \sqrt{\frac{6.59}{9.81}} = 9.1 \text{ s}$$

$$N = \frac{10800s}{T_{S,1}} = \frac{10800}{9.1} = 1187 \Rightarrow H_{m,1} = 12.4 \text{ m}$$

$$T_{m,1} = 11.1 \sqrt{\frac{12.4}{9.81}} = 12.48 \text{ s}$$

For the 50-year extreme wave height:

$$H_{m,50} = H_{S,50} \sqrt{\frac{1}{2} \ln(N)}$$

where,

$$N = \frac{10800s}{10.97} = \frac{10800}{10.97} = 982 \Rightarrow H_{m,50} = 15.33 \text{ m}$$

$$T_{m,50} = 11.1 \sqrt{\frac{15.33}{9.8}} = 13.88 \text{ s}$$

The wave periods and associated parameters from DNVGL standards

are applied within the Airy wave theory framework. The time-dependent hydrodynamic loads on the structural components are calculated using Eqs. (39)–(45), incorporating wave heights and periods from prior analysis. The structural loads are determined through vector summation of forces and moments to obtain maximum horizontal loads on the support structure. Then, the resultant forces are multiplied by the dynamic amplification factor (DAF) to account for structural dynamic responses.

For the 1-year extreme wave height:

$$DAF = \frac{1}{\sqrt{\left(1 - \left(\frac{f}{f_0}\right)^2\right)^2 + \left(2\xi \frac{f}{f_0}\right)^2}} = \frac{1}{\sqrt{(1 - 0.3^2)^2 + (0.1 \times 0.3)^2}} = 1.08$$

For the 50-year extreme wave height:

$$DAF = \frac{1}{\sqrt{\left(1 - \left(\frac{f}{f_0}\right)^2\right)^2 + \left(2\xi \frac{f}{f_0}\right)^2}} = \frac{1}{\sqrt{(1 - 3.7^2)^2 + (0.1 \times 3.7)^2}} = 1.06$$

The jacket structure is analyzed using the layer-by-layer computational approach from Subsection 2.2. For the layer 1, the maximum horizontal load is:

$$H = \frac{F_{aero} + F_{hydro}}{4} = 0.87 \text{ MN}$$

The equivalent compressive force acting in braces is determined from static equilibrium:

$$F_b = \frac{H}{\cos \theta_h} = \frac{0.87}{\cos 62.1^\circ \times \cos 45^\circ} = 2.62 \text{ MN}$$

The allowable compressive stress for tubular members is calculated using Eq. (53):

$$\sigma_{allow} = \frac{12\pi^2 E}{23 \times \left(\frac{KL}{r}\right)^2} = \frac{12\pi^2 \times 2.1 \times 10^5}{23 \times 166^2} = 40.3 \text{ MPa}$$

$$\text{where } f_y = 355, C_c = \left(\frac{2\pi^2 E}{f_y}\right)^{\frac{1}{2}} = 108.05.$$

The axial compressive load capacity of the brace is:

$$F = \frac{\sigma_{allow} A_b}{\gamma_f} = \frac{40.3 \times 0.014}{1.15} = 0.48 \text{ MN}$$

where, γ_f is material safety factor (taken as 1.15). The allowable load is below the design load (2.62 MN).

The structural analysis proceeded with verification of the maximum compressive force in the legs. For the critical load case where aerodynamic and hydrodynamic loads act at 45° to the x-axis, the governing load case for legs is established. Hydrodynamic load calculations are recalibrated based on the updated directional components, incorporating buoyancy forces from submerged elements and applying a safety factor of 1.3 to the structural self-weight. For tension members, a favorable safety factor of 0.9 is applied to the structural self-weight. The axial forces in the legs, comprising net weight (with buoyancy adjustment) combined with transverse wind and wave effects, are computed as follows:

$$F_{N1} = \frac{1}{8.85} \left(\frac{247.6}{\sqrt{2}}\right) + \frac{1.3 \times 12.9}{4} = 24 \text{ MN}$$

$$F_{N4} = \frac{1}{8.85} \left(\frac{247.6}{\sqrt{2}}\right) - \frac{0.9 \times 12.9}{4} = 31.9 \text{ MN}$$

The axial compressive strength of jacket legs is calculated using the

Table 5

Summary of applied loads.

Aero Load		Hydro Load ()	
$F_{aero.ETM}$	1.03 MN	$F_{hydro.ETM}$ (Including DAF)	0.98 MN
$F_{aero.EOG}$	1.74 MN	$F_{hydro.EOG}$ (Including DAF)	0.98 MN
$F_{aero.EWM6.1}$	2.25 MN	$F_{hydro.EWM6.1}$ (Including DAF)	3.85 MN
$F_{aero.EWM6.3}$	1.44 MN	$F_{hydro.EWM6.3}$ (Including DAF)	2.67 MN
$M_{aero.ETM}$	140.9 MN m	$M_{hydro.ETM}$ (Including DAF)	64.0 MN m
$M_{aero.EOG}$	238.1 MN m	$M_{hydro.EOG}$ (Including DAF)	64.0 MN m
$M_{aero.EWM6.1}$	307.7 MN m	$M_{hydro.EWM6.1}$ (Including DAF)	166.8 MN m
$M_{aero.EWM6.3}$	196.9 MN m	$M_{hydro.EWM6.3}$ (Including DAF)	116.4 MN m

Table 6

Summary of results.

Parameters	Proposed method		UpWind Jacket
Jacket Height	67 m		67 m
Jacket Leg Section	Layer 1	1200 × 35	1200 × 35
	Layer 2	1200 × 40	1200 × 40
	Layer 3	1200 × 50	1200 × 40
	Layer 4	1200 × 60	1200 × 50
Jacket Brace Section	Layer 1	600 × 30	800 × 20
	Layer 2	600 × 30	
	Layer 3	600 × 30	
	Layer 4	600 × 30	

method similar with braces. The effective length factor μ is taken as 1.0 in this analysis.

$$\sigma_{allow} = \frac{\left(1 - \left(\frac{68.1}{\sqrt{2 \times 108.05}}\right)^2\right) \times 355}{\frac{5}{3} + \frac{3 \times 68.1}{8 \times 108.05} - \frac{1}{8} \left(\frac{68.1}{108.05}\right)} = 152.1 \text{ MPa}$$

The total axial compressive capacity of the jacket leg is:

$$F = \frac{\sigma_{allow} A_L}{\gamma_f} = \frac{152.1 \times 0.046}{1.15} = 6.1 \text{ MN}$$

The members of layer 1 required dimensional increases as the applied loads exceeded their axial compressive capacity. Following the methodology in Subsection 2.2, tubular members are redesigned by gradually increasing diameters and wall thicknesses, with iterative verification of sectional capacity until compliance is achieved. Subsequent layers are designed using a consistent framework: a top-to-down layer-by-layer methodology is implemented, where each layer underwent identical load analysis, member optimization, and stability validation processes, ensuring full compliance with structural performance requirements across the entire jacket system.

Table 5 summarizes the total structural loads acting on layer 4 of the jacket under the specified load cases. Table 6 summarizes the dimensional parameters of the jacket structure designed using the proposed method with those from Ref. [70]. The results of the simplified design are found to be acceptable.

3.1.4. Determine the pile dimension

The stratigraphic profile of the site predominantly consists of sandy soil, with geotechnical properties as detailed in ref. [73]. According to this report, the composite effective internal friction angle (ϕ') for the sandy stratum is 35°. Based on the subsoil stratification and bearing capacity characteristics, the pile length is determined as 50 m, and the pile diameter is initialized to 2 m.

For the pile design, a material partial factor of 1.15 and a global stability factor of 1.25 were adopted. Based on the API-specified shaft friction resistance theory (API, 2000), the minimum pile diameter is determined by iteratively increasing the pile diameter and performing stratified layer-by-layer calculation of shaft friction resistance. The iterative process continues until the total bearing capacity satisfies the

design load, yielding the optimal pile diameter, D_{pile} :

$$D_{pile} = 2.3 \text{ m}$$

The pile thickness, t_{pile} , is calculated according to API recommendations (API, 2000):

$$t_{pile} = 6.35 + \frac{2.3}{100} = 0.03 \text{ m}$$

3.1.5. Computing natural frequency and tower-top displacement

Based on dimensions of the jacket and pile foundation established previously, the fundamental frequency of the 5 MW JOWT is analyzed using the methodology provided in Subsection 2.7. The equivalent stiffness of the jacket, EI_J , and the fundamental frequency of the system, f_0 , are as follows:

$$EI_J = 6.28 \times 10^{12} \text{ N}\cdot\text{m}^2$$

$$f_0 = 0.291$$

The structural deformation complies with the serviceability limit state criteria. The aerodynamic load under the running condition of OWTs, calculated using the normal turbulence model, is 0.83 MN. For extreme sea states with a 1-year return period, the hydrodynamic load is determined as 2.47 MN, with a corresponding foundation-applied moment of 107.7 MN·m. The eccentricity of the wave load is calculated as $\frac{107.7}{2.47} = 43.6 \text{ m}$. Consequently, the effective structural height, a , and tower-top deflection induced by hydrodynamic loadings, δ_{hydro} , are derived as:

$$a = 137 - 43.6 = 93.4 \text{ m}$$

$$\delta_{hydro} = \frac{F_{hydro} h_{total} (h_{total} - a)}{k_R} + \frac{F_{hydro} h_{total}^3}{3EI_J} \left[\frac{(h_{total}^2 - a^2)}{3} - \frac{a(h_{total}^2 - a^2)}{2} \right]$$

$$= 0.085 \text{ m}$$

The deformation of the structure under aerodynamic load is calculated using the system's equivalent stiffness, EI_{T-J} :

$$\delta_{aero} = \frac{F_{aero} h_{total}^2}{k_R} + \frac{F_{aero} h_{total}^3}{3EI_{T-J}} = 0.49 \text{ m}$$

The total displacement at the tower top is:

$$\delta_{total} = \delta_{hydro} + \delta_{aero} = 0.575 \text{ m}$$

It is noteworthy that the displacement verification methodology proposed in this study establishes a preliminary safety assessment framework for allowable wind turbine foundation deformations. The method employs a conservative equivalent concentrated load model for hydrodynamic load computation and deliberately neglects the shear stiffness effects of the support structure, thereby ensuring conservative safety margins in the evaluation results. This model is grounded in simplified linear-elastic subgrade reaction hypotheses; subsequent refinements will incorporate nonlinear SSI mechanisms to enhance

Table 7

Design parameters for 10 MW offshore wind turbine.

Tower	Value
Tower Height, h_T (m)	89
Top Diameter, D_{top} (m)	5.5
Bottom Diameter, D_{bottom} (m)	7.6
Equivalent Tower Wall Thickness, t_T (m)	0.0299
Total Tower Mass, m_T (kg)	430000
Mass of RNA, M_{RNA} (kg)	676738
Site Water Depth, S (m)	50
50-Year characteristic Wave Height, $H_{s,50}$ (m)	9.4 m
50-year extreme wave height, $H_{m,50}$ (m)	17.2 m
The extreme wind speed with a 50-year return period, $U_{e,50}$ (m/s)	42.73
The extreme wind speed with a 1-year return period, $U_{e,1}$ (m/s)	34.18

Table 8

Summary of external dimensions for each layer of the 10 mw jacket structure.

Layer number	Height (m)	Width (m)
1	12.03	17.47
2	14.74	21.81
3	18.07	27.22
4	22.16	34

Table 9

Summary of applied loads.

Aero Load		Hydro Load	
$F_{aero,ETM}$	2.12 MN	$F_{hydro,ETM}$ (Including DAF)	1.45 MN
$F_{aero,EOG}$	3.27 MN	$F_{hydro,EOG}$ (Including DAF)	1.45 MN
$F_{aero,EWM6.1}$	4.50 MN	$F_{hydro,EWM6.1}$ (Including DAF)	5.23 MN
$F_{aero,EWM6.3}$	2.88 MN	$F_{hydro,EWM6.3}$ (Including DAF)	3.45 MN
$M_{aero,ETM}$	357.3 MN m	$M_{hydro,ETM}$ (Including DAF)	75.8 MN m
$M_{aero,EOG}$	551.1 MN m	$M_{hydro,EOG}$ (Including DAF)	75.8 MN m
$M_{aero,EWM6.1}$	759.5 MN m	$M_{hydro,EWM6.1}$ (Including DAF)	227.8 MN m
$M_{aero,EWM6.3}$	486.1 MN m	$M_{hydro,EWM6.3}$ (Including DAF)	149.2 MN m

Table 10

Summary of results.

	Proposed method		INNWind.EU
Jacket height	67 m		67 m
Jacket Leg Section	Layer 1	1400 × 45	1400 × 42
	Layer 2	1400 × 50	1400 × 42
	Layer 3	1400 × 60	1400 × 48
	Layer 4	1400 × 70	1400 × 60
Jacket Brace Section	Layer 1	850 × 20	832 × 16
	Layer 2	850 × 25	840 × 20
	Layer 3	850 × 30	936 × 18
	Layer 4	850 × 40	1060 × 30

computational fidelity.

3.2. Preliminary design of jacket support structure for 10 MW OWT

The preliminary design methodology for the 10 MW JOWT in this section follows the same computational procedures as those applied to the 5 MW JOWT design. To avoid redundancy, the detailed computational workflow is not reiterated here; only key results are presented. The fundamental design parameters for jacket of this 10 MW OWT are summarized in Table 7, with detailed technical specifications available in Ref. [74].

3.2.1. Overall geometric dimensions of jacket

Based on the parameters in Table 7 and Eq. (22), the height of the jacket is set as 67 m. Additional external dimensions of the jacket for the 10 MW JOWT are provided in Table 8.

3.2.2. Determine the cross-sections of legs and braces

Table 9 summarizes the total structural loads acting on layer 4 of the jacket under different load cases.

Table 10 compares the dimensional parameters of the 10 MW OWT jacket components designed using the proposed methodology with those from Ref. [71]. The comparison demonstrates that the simplified design methodology is reliable.

3.2.3. Dimension of pile, system frequency and tower-top displacement

Based on engineering practice, the outer diameter of the pile, D_{pile} , is set to 2.8 m. Through iterative calculations based on the previously determined loads, the pile length, L_{pile} , is determined as 40 m, with a wall thickness, t_{pile} , of 0.035 m. Subsequently, the calculated natural frequency of the system considering SSI, f_0 , is 0.2873 Hz. The tower-top

displacement under serviceability limit state load cases, δ_{total} , is 0.583 m.

4. Conclusions and outlooks

To address the inherent complexity in jacket and foundation design for OWTs, this study develops a novel integrated methodology. This method features a variable-cross-section jacket design method coupled with equivalent stiffness models for dynamic analysis. Key innovations include: (1) a top-to-down, layered design methodology incorporating hydrodynamic loads, enabling layer-specific member sizing with high computational efficiency (typically achieved within 10 min after data collection); (2) derivation of equivalent stiffness for tapered jackets based on equivalent horizontal displacement at the transition piece, enabling rapid assessment of natural frequency and tower-top deflection of the system; and (3) development of an open-source MATLAB computational tool implementing this methodology, publicly available on GitHub. This methodology is validated through comparison with jacket designs of the Upwind and INNWind projects, yielding the following conclusions:

1. A tapered jacket model, represented as an equivalent uniform cross-section Euler-Bernoulli beam, is proposed for fundamental frequencies analysis of JOWTs. The validation against high-fidelity finite element models confirms that the proposed frequency calculation method for JOWTs, based on equivalent stiffness of variable-section jackets, achieves engineering-acceptable accuracy (error <2.5%).
2. The numerical analysis demonstrates that the Ultimate Limit State (ULS) exerts substantially stronger control over jacket member cross-sectional dimensions than frequency constraints. This confirms that structural strength indices (e.g., compressive resistance of members) must be prioritized as primary design criteria. In ULS analysis, dominant load combinations should be identified first, with focused verification of their mechanical impacts on critical sections.
3. The proposed method yields cross-sectional dimensions for specific members that are 10–20% larger than those in benchmark designs, yet generates the preliminary design within 10 min once the necessary data was collected. This demonstrates its efficacy in preliminary design stages of jacket foundations, satisfying ULS and frequency constraints while retaining potential for further optimization.

While this study establishes an efficient framework for the preliminary design of jacket support structures, several limitations constrain its scope and necessitate future research. Firstly, the reliance on static analysis with a Dynamic Amplification Factor and linear regular wave theory for load calculation prioritizes computational efficiency but does not fully capture stochastic wave dynamics or coupled system behavior. The future work should thus implement fully coupled time-domain dynamic analysis employing irregular wave theory for more accurate load prediction. Secondly, the framework addresses Ultimate and Serviceability Limit States but currently excludes systematic Fatigue Limit State assessment, which is a critical design driver of OWTs. The comprehensive fatigue analysis on the preliminary designs generated by using this method is therefore an essential next step. Finally, while dominant wind and wave loads are considered, seismic and tropical cyclone (typhoon/hurricane) loads are not integrated into the core iterative design loop. Future efforts should develop tools to tightly couple integrated dynamic analysis modules for these specific hazards, enabling a more comprehensive multi-hazard design framework.

CRedit authorship contribution statement

Renqiang Xi: Funding acquisition, Data curation, Conceptualization. **Haipeng Shan:** Writing – original draft, Methodology, Data curation. **Meiling Cheng:** Software, Project administration, Formal analysis.

M. Hesham El Naggar: Writing – review & editing, Investigation, Conceptualization. **Xiuli Du:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A

$$T_1 = \frac{h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{(h_3+h_4)(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{(h_3+h_4)(L_b-L_t)}{h_J}} \right)}{Ac_2 \cdot E_J \cdot (L_b - L_t)^2} \quad (A1)$$

$$T_2 = - \frac{h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{(h_3+h_4)(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{(h_3+h_4)(L_b-L_t)}{h_J}} \right)}{Ac_3 \cdot E_J \cdot (L_b - L_t)^2} \quad (A2)$$

$$T_3 = \frac{3h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_J}} \right)}{Ac_1 \cdot E_J \cdot (L_b - L_t)^2} \quad (A3)$$

$$T_4 = - \frac{3h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_J}} \right)}{Ac_2 \cdot E_J \cdot (L_b - L_t)^2} \quad (A4)$$

$$T_5 = 3h_4 \quad (A5)$$

$$T_6 = -3h_4 \quad (A6)$$

$$T_7 = T_{7a} + T_{7b} + T_{7c} + T_{7d} + T_{7f} + T_{7g} \quad (A7)$$

$$T_{7a} = \frac{h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{(h_3+h_4)(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{(h_3+h_4)(L_b-L_t)}{h_J}} \right)}{Ac_2 \cdot E_J \cdot (L_b - L_t)^2} \quad (A8)$$

$$T_{7b} = - \frac{h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{(h_3+h_4)(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{(h_3+h_4)(L_b-L_t)}{h_J}} \right)}{Ac_3 \cdot E_J \cdot (L_b - L_t)^2} \quad (A9)$$

$$T_{7c} = \frac{h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{h_4(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{h_4(L_b-L_t)}{h_J}} \right)}{Ac_3 \cdot E_J \cdot (L_b - L_t)^2} \quad (A10)$$

$$T_{7d} = - \frac{h_J^2 \cdot \left(\ln \left(\left| L_b - \frac{h_4(L_b-L_t)}{h_J} \right| \right) + \frac{L_t}{L_b \cdot \frac{h_4(L_b-L_t)}{h_J}} \right)}{Ac_4 \cdot E_J \cdot (L_b - L_t)^2} \quad (A11)$$

$$T_{7e} = \frac{h_j^2 \cdot \left(\ln L_b + \frac{L_t}{L_b} \right)}{Ac_4 \cdot E_J \cdot (L_b - L_t)^2} \quad (A12)$$

$$T_{7f} = \frac{h_j^2 \cdot \left(\ln \left(L_b - \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_j} \right) + \frac{L_t}{L_b - \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_j}} \right)}{Ac_1 \cdot E_J \cdot (L_b - L_t)^2} \quad (A13)$$

$$T_{7g} = - \frac{h_j^2 \cdot \left(\ln \left(L_b - \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_j} \right) + \frac{L_t}{L_b - \frac{(h_2+h_3+h_4)(L_b-L_t)}{h_j}} \right)}{Ac_2 \cdot E_J \cdot (L_b - L_t)^2} \quad (A14)$$

$$T_8 = \frac{3h_j^2 \cdot \left(\ln \left(L_b - \frac{(h_3+h_4)(L_b-L_t)}{h_j} \right) \cdot \left(L_b + L_t - \frac{(h_3+h_4)(L_b-L_t)}{h_j} \right) - L_b + \frac{(h_3+h_4)(L_b-L_t)}{h_j} \right)}{Ac_2 \cdot E_J \cdot (L_b - L_t)^3} \quad (A15)$$

$$T_9 = \frac{3h_j^2 \cdot \left(\ln \left(L_b - \frac{(h_3+h_4)(L_b-L_t)}{h_j} \right) \cdot \left(L_b + L_t - \frac{(h_3+h_4)(L_b-L_t)}{h_j} \right) - L_b + \frac{(h_3+h_4)(L_b-L_t)}{h_j} \right)}{Ac_3 \cdot E_J \cdot (L_b - L_t)^3} \quad (A16)$$

Data availability

I have shared the code in a Github link.

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